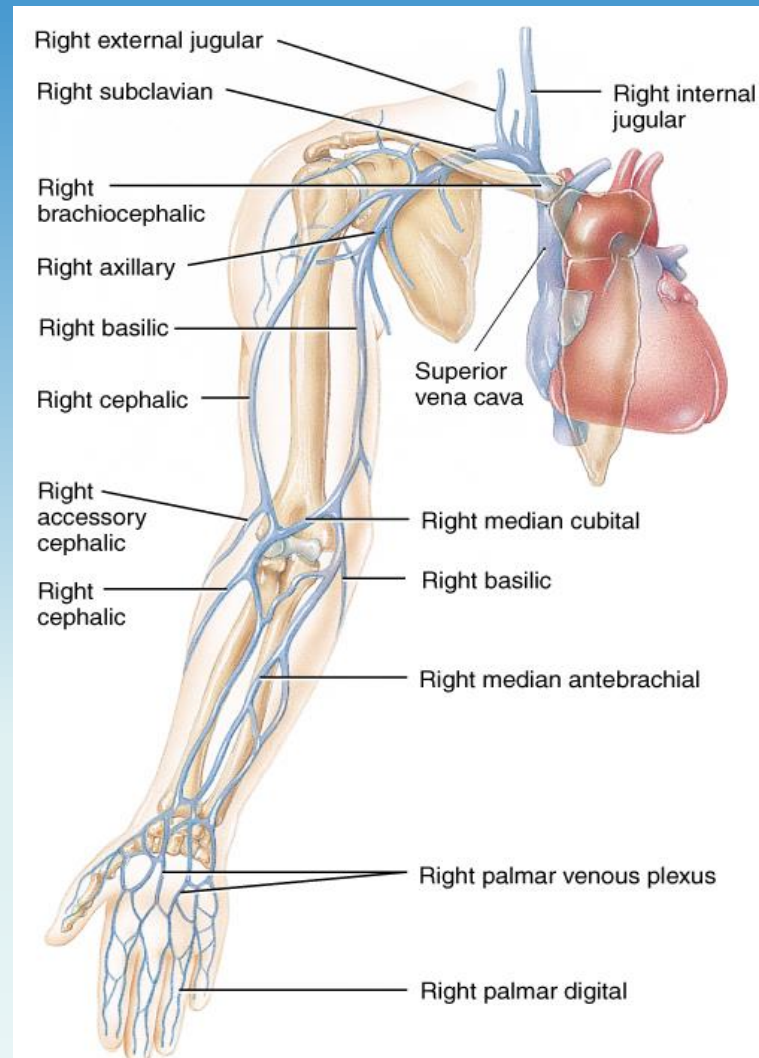


# The Cardiovascular System: Blood Vessels and Hemodynamics



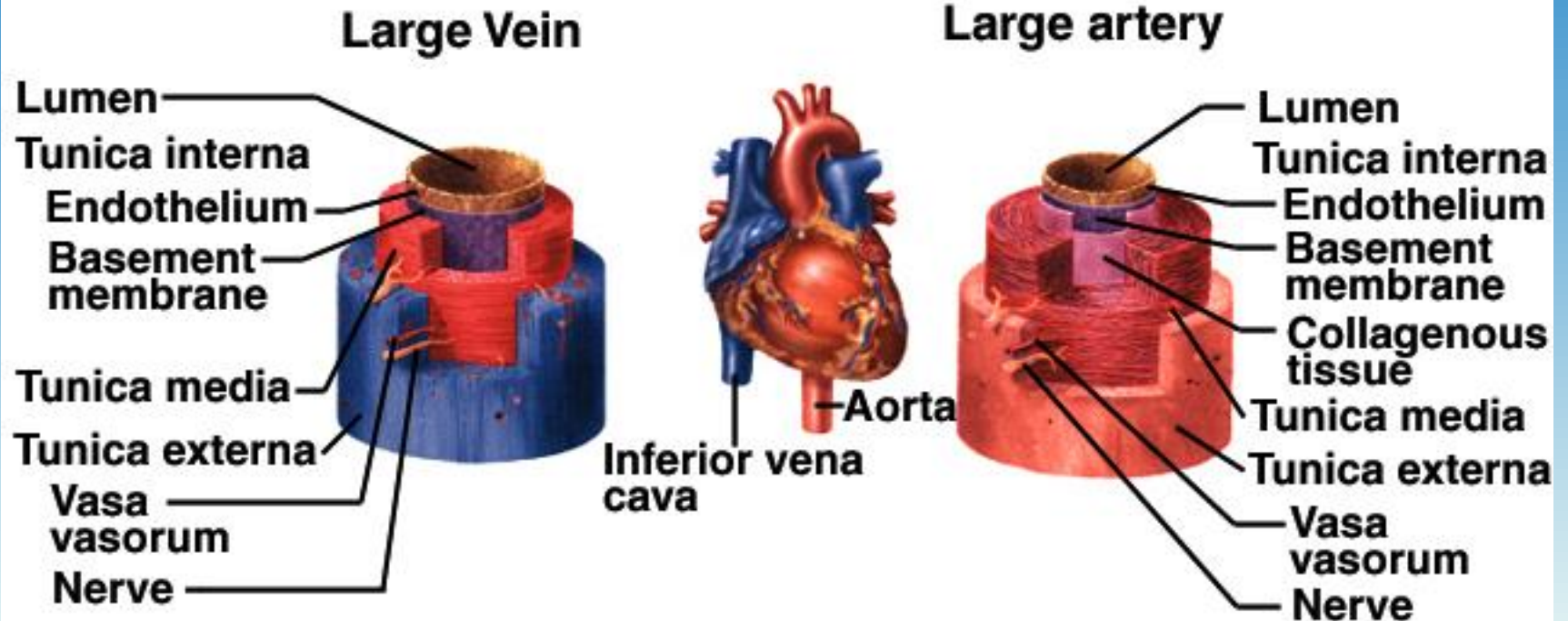
# Anatomy of Blood Vessels

- Closed system of tubes that carries blood
- Arteries carry blood from heart to tissues
  - elastic arteries
  - muscular arteries
  - arterioles
- Capillaries are thin enough to allow exchange
- Venules merge to form veins that bring blood back to the heart
- Vasa vasorum is vessels in walls of large vessel

# The Vessel Wall

- Tunica externa
  - outermost layer
  - loose connective tissue
- Tunica media
  - middle layer
  - usually thickest; smooth muscle, collagen, some elastic
  - smooth muscle for vasoconstriction and vasodilation
- Tunica intima (interna)
  - smooth inner layer that repels blood cells & platelets
  - simple squamous endothelium overlying a basement membrane and layer of fibrous tissue

# Large Vessels



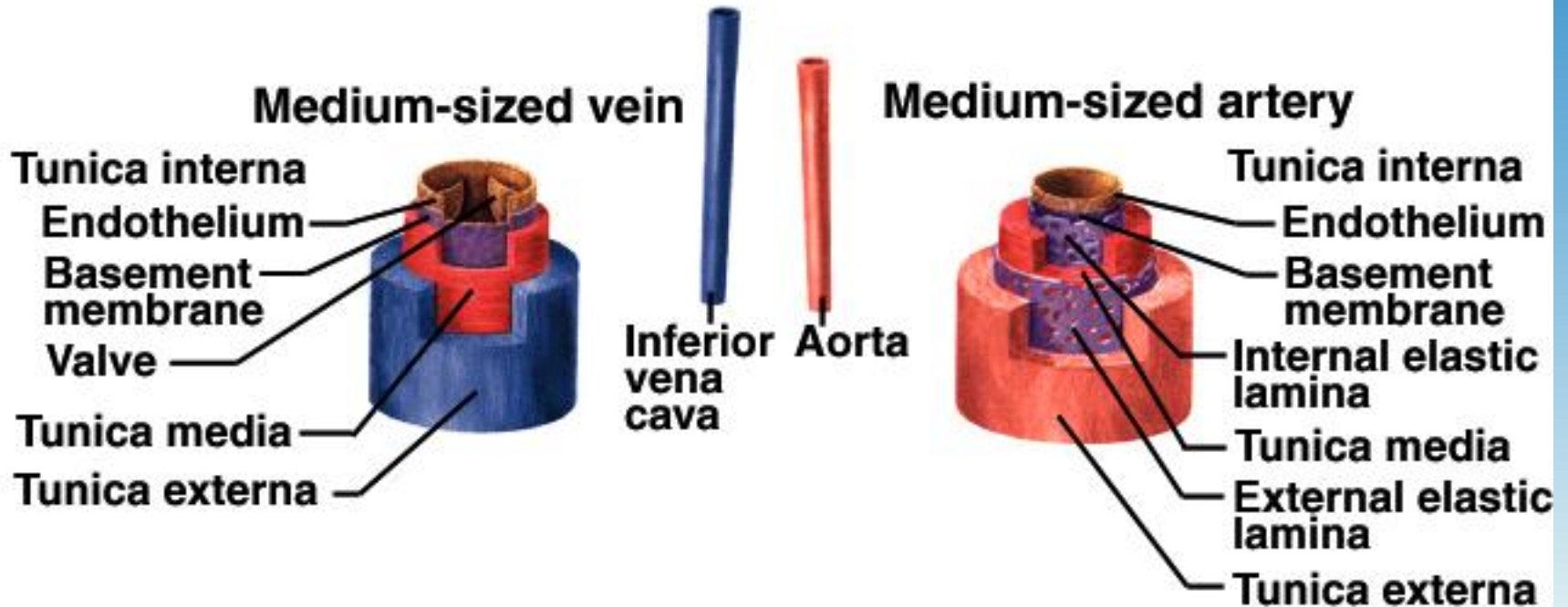
# Sympathetic Innervation

- Vascular smooth muscle is innervated by sympathetic nervous system
  - increase in stimulation causes muscle contraction or vasoconstriction
    - decreases diameter of vessel
  - injury to artery or arteriole causes muscle contraction reducing blood loss (vasospasm)
  - decrease in stimulation or presence of certain chemicals causes vasodilation
    - increases diameter of vessel
    - nitric oxide,  $K^+$ ,  $H^+$  and lactic acid cause vasodilation

# Arteries

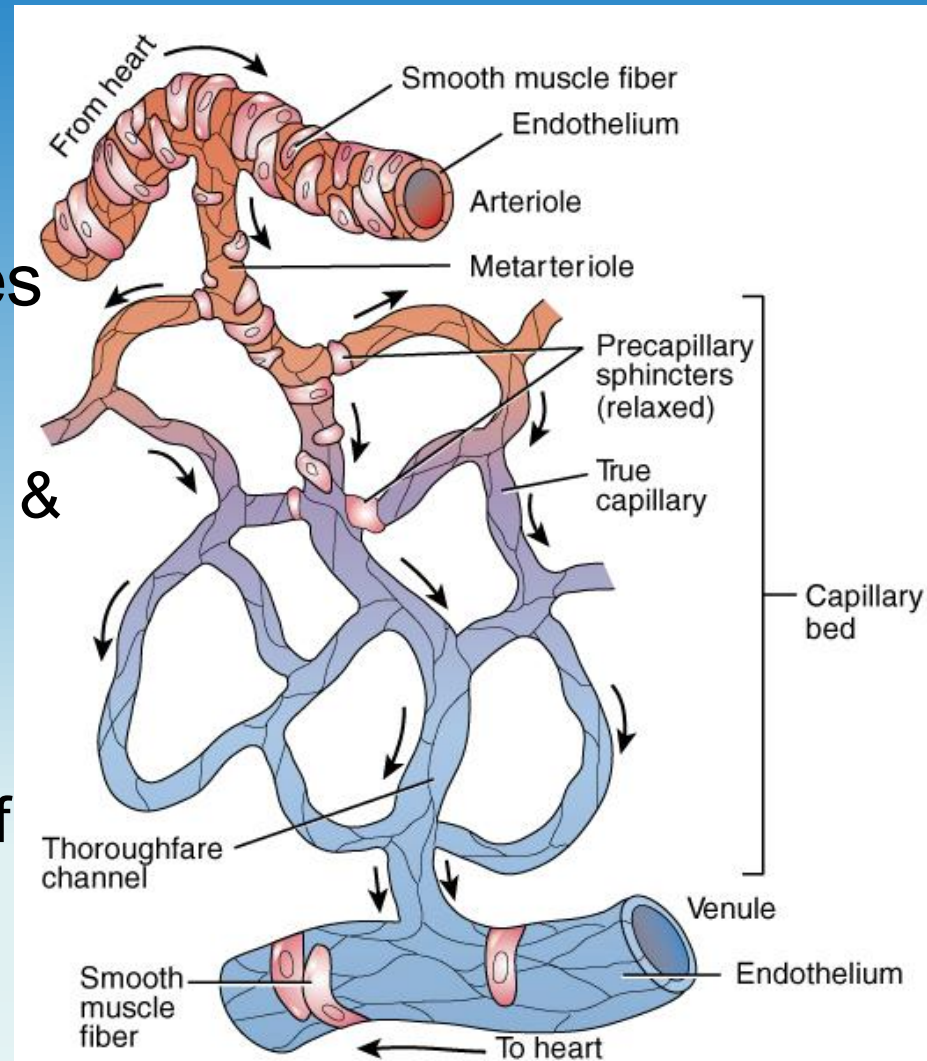
- Conducting (elastic) arteries - largest
  - pulmonary, aorta and common carotid
  - tunica media consists of perforated sheets of elastic tissue, alternating with thin layers of smooth muscle, collagen and elastic fibers
  - expand during systole, recoil during diastole; lessens fluctuations in BP
- Distributing (muscular) arteries
  - distributes blood to specific organs; femoral and splenic
  - smooth muscle layers constitute  $\frac{3}{4}$  of wall thickness

# Medium Vessels

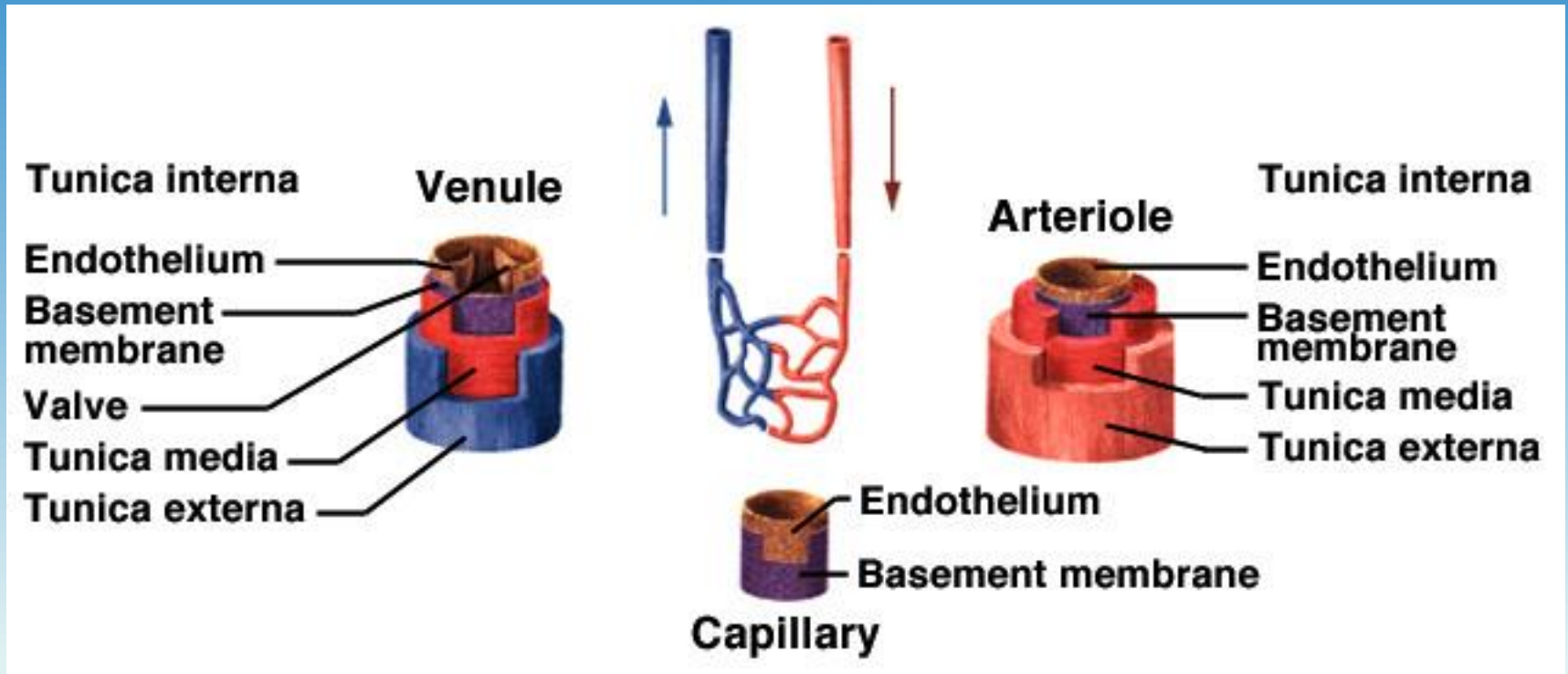


# Arterioles

- Small arteries delivering blood to capillaries
  - tunica media containing few layers of muscle
- Metarterioles form branches into capillary bed
  - to bypass capillary bed, precapillary sphincters close & blood flows out of bed in thoroughfare channel
  - vasomotion is intermittent contraction & relaxation of sphincters that allow filling of capillary bed 5-10 times/minute

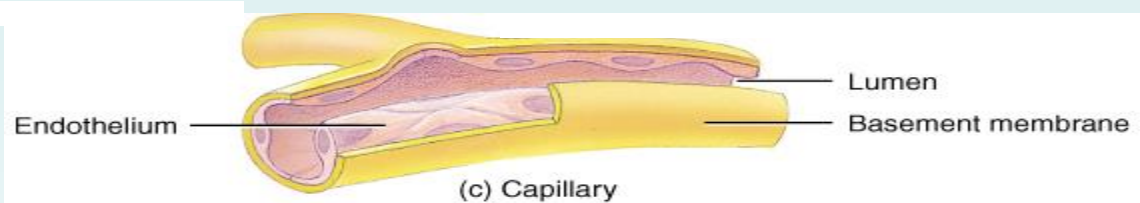


# Small Vessels

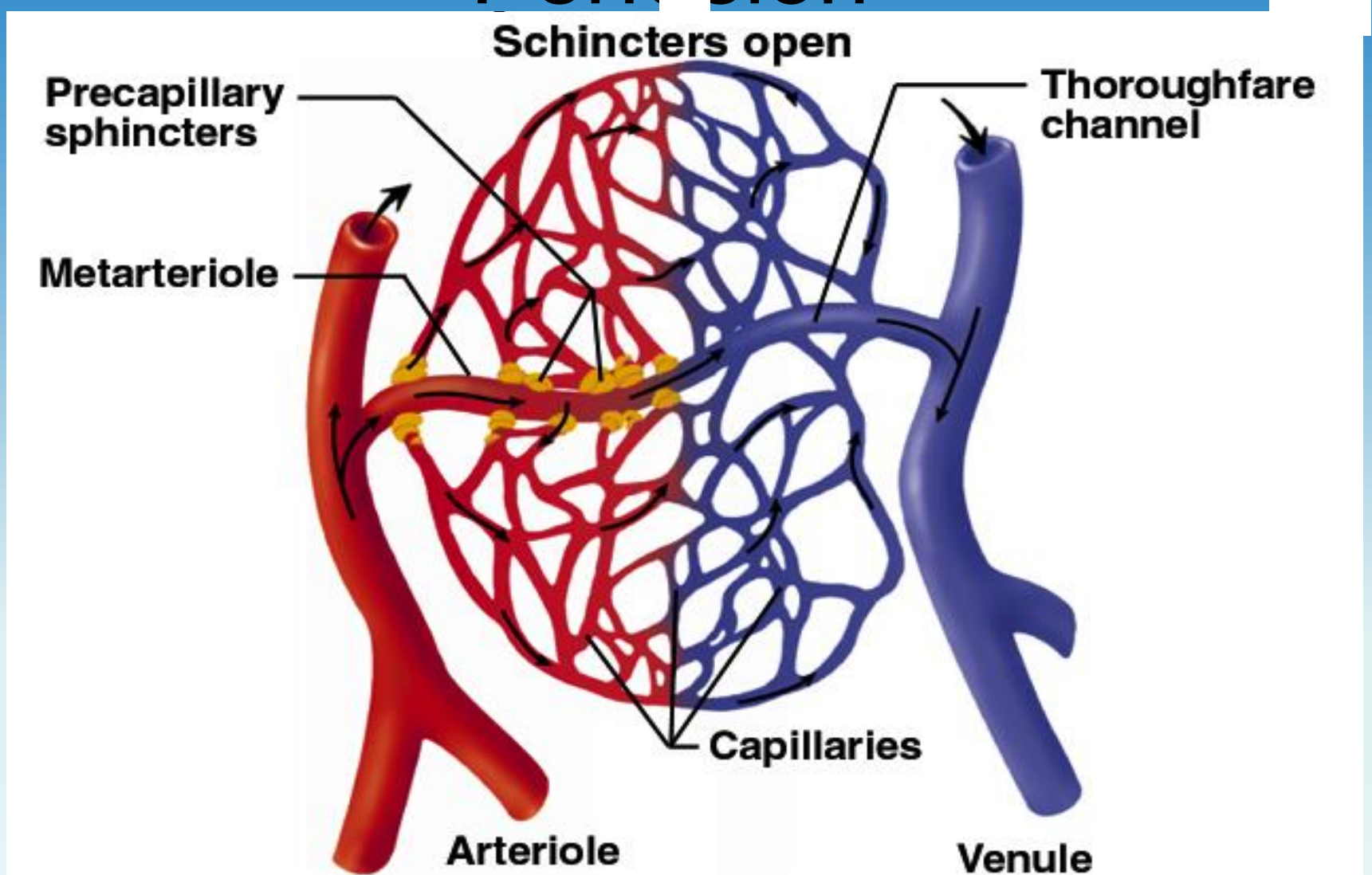


# Capillaries form Microcirculation

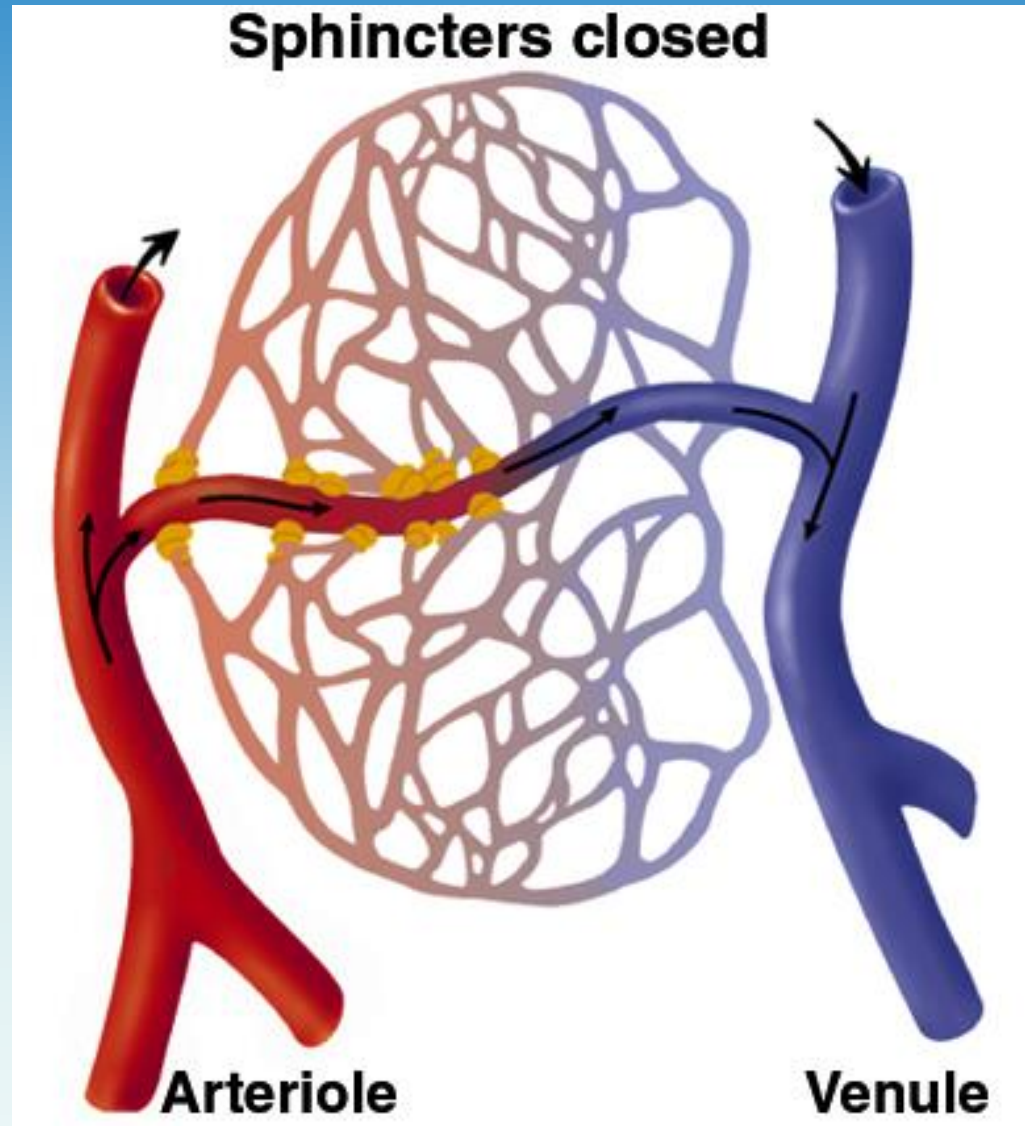
- Microscopic vessels that connect arterioles to venules
- Found near every cell in the body but more extensive in highly active tissue (muscles, liver, kidneys & brain)
  - entire capillary bed fills with blood when tissue is active
  - lacking in epithelia, cornea and lens of eye & cartilage
- Function is exchange of nutrients & wastes between blood and tissue fluid
- Structure is single layer of simple squamous epithelium and its basement membrane



# Control of Capillary Bed Perfusion



# Control of Capillary Bed Perfusion



# Types of Capillaries

- Continuous capillaries

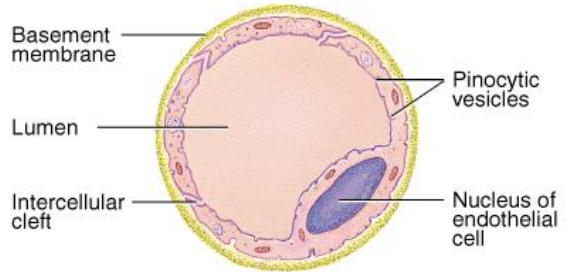
- intercellular clefts are gaps between neighboring cells
- skeletal & smooth, connective tissue and lungs

- Fenestrated capillaries

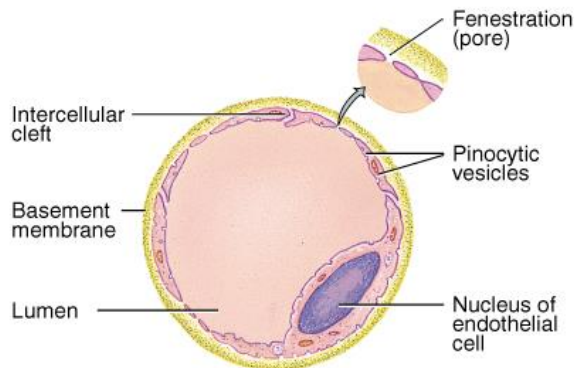
- plasma membranes have many holes
- kidneys, small intestine, choroid plexuses, ciliary process & endocrine glands

- Sinusoids

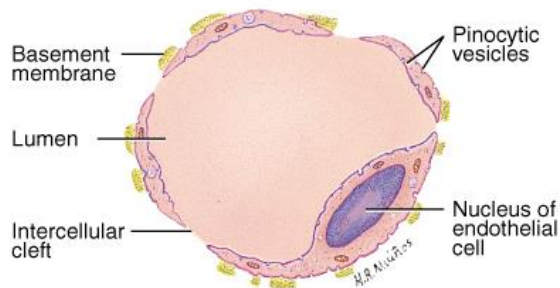
- very large fenestrations
- incomplete basement membrane
- liver, bone marrow, spleen, anterior pituitary, & parathyroid gland



(a) Continuous capillary formed by endothelial cells

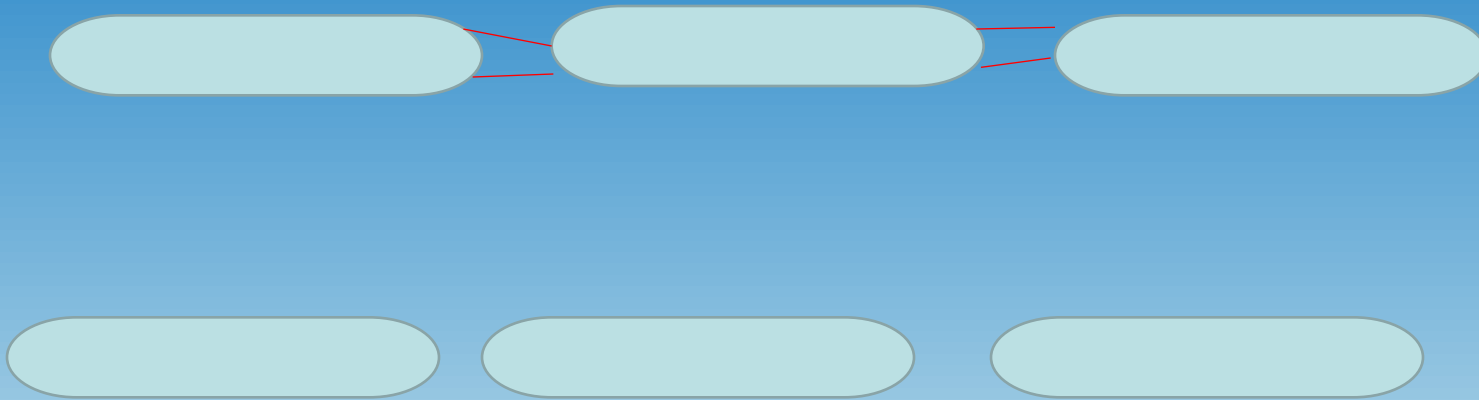


(b) Fenestrated capillary

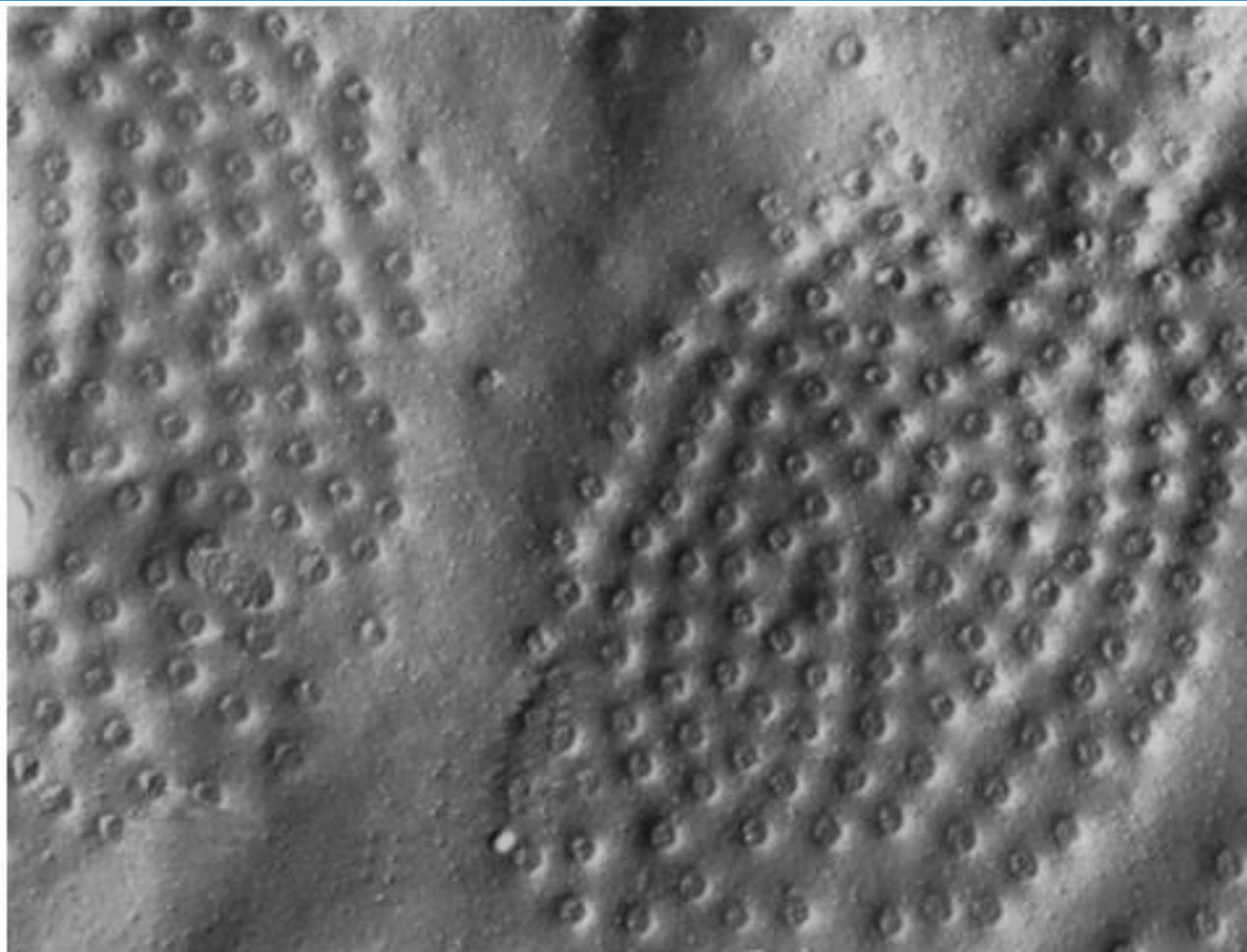


(c) Sinusoid

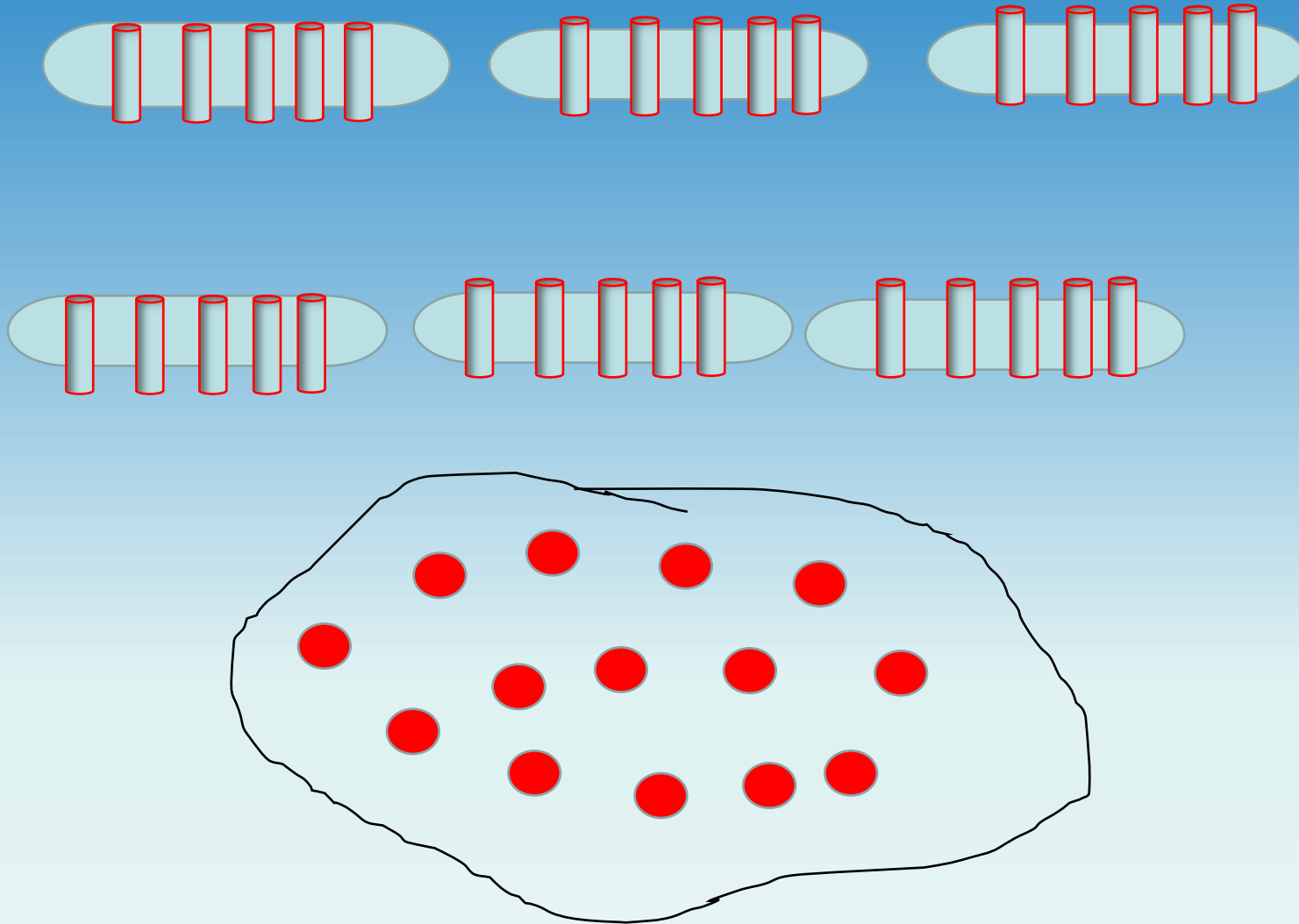
# Continuous capillary



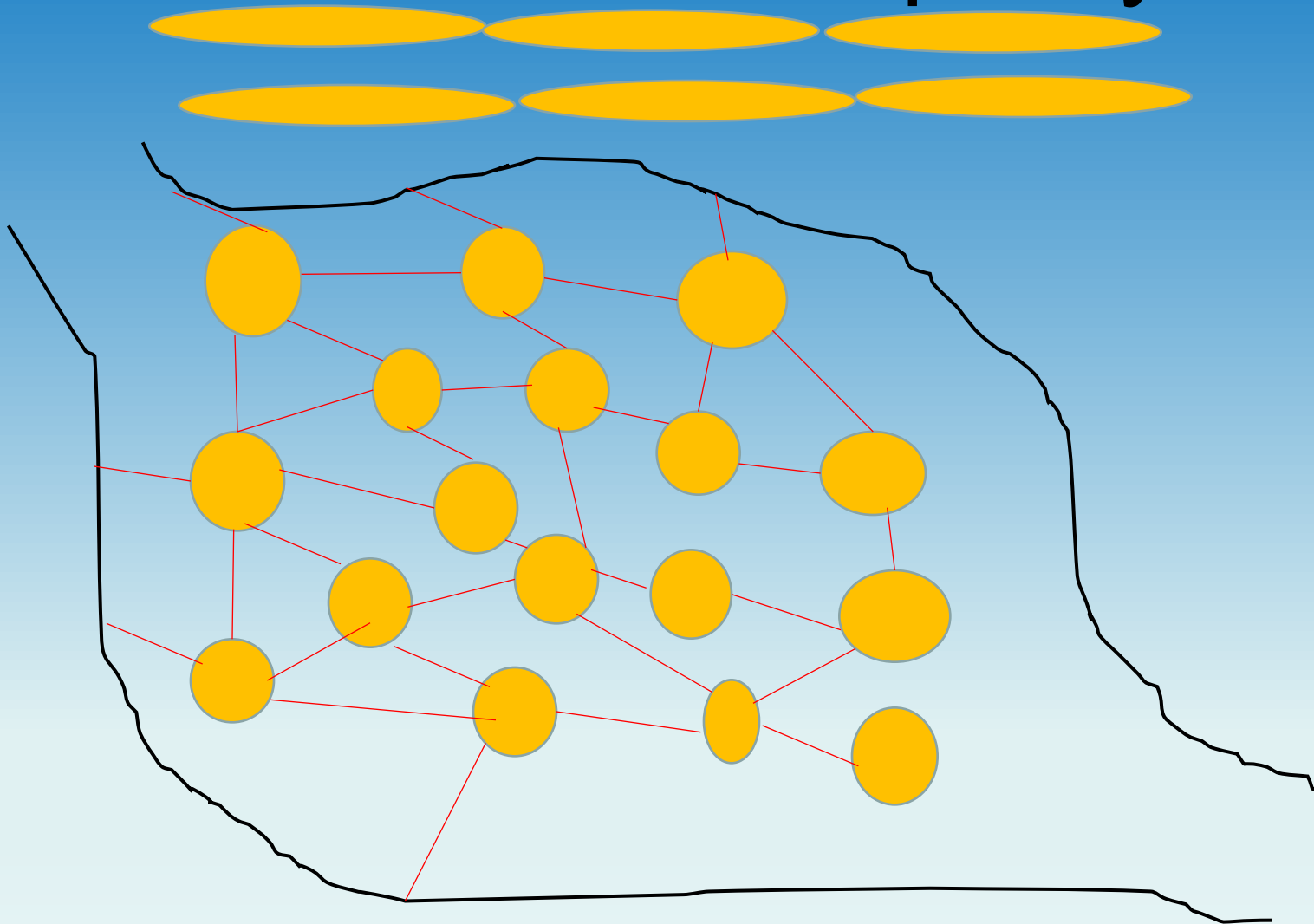
# Fenestrated Endothelial Cell



# Fenestrated capillary



# Sinusoidal capillary



# Veins

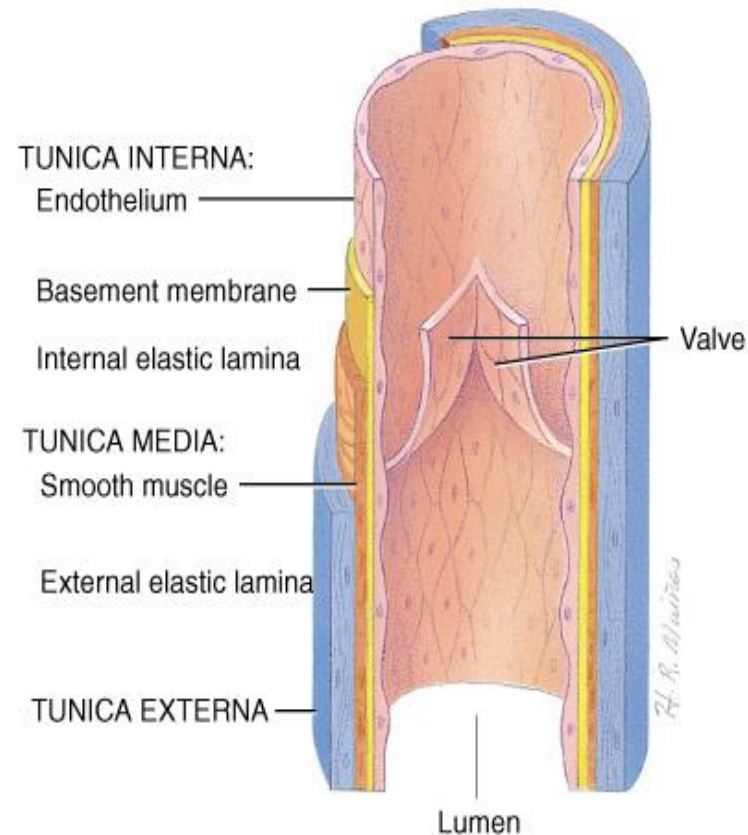
- Venules
  - proximal venule is quite porous, exchanges fluid with tissues, like a capillary, at this point only
- Venous sinuses: veins with thin walls, large lumens, no smooth muscle
- Veins have lower blood pressure: avg.. 10mmHg with little fluctuation
  - thinner walls, less muscular and elastic tissue
  - expand easily, have high capacitance
  - venous valves aid skeletal muscles in upward blood flow

# Venules

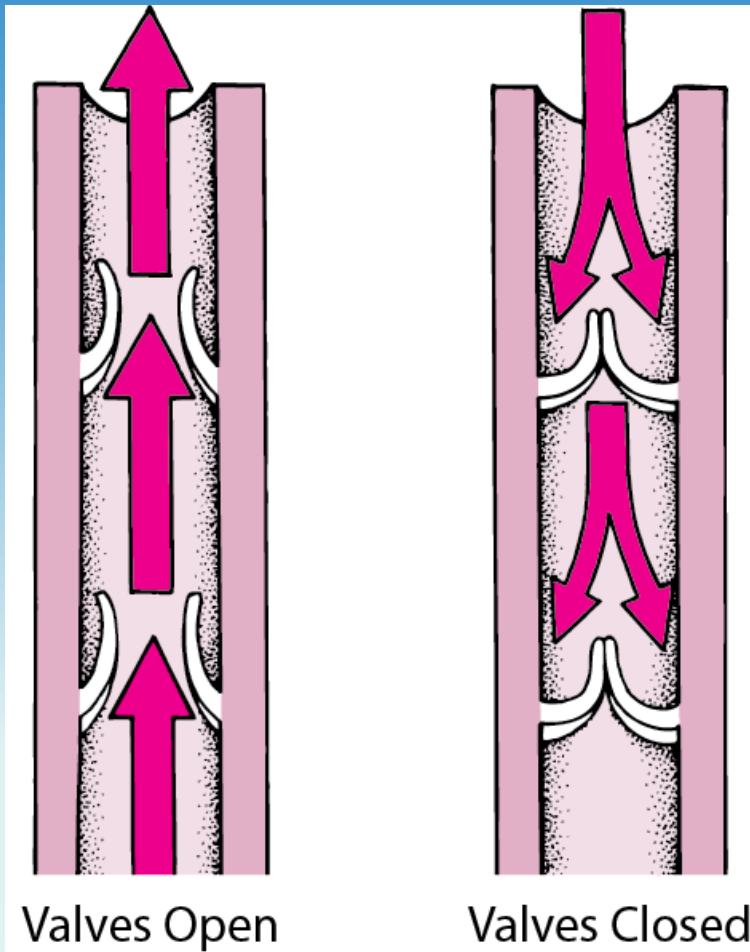
- Small veins collecting blood from capillaries
- Tunica media contains only a few smooth muscle cells & scattered fibroblasts
  - very porous endothelium allows for escape of many phagocytic white blood cells
- Venules that approach size of veins more closely resemble structure of vein

# Veins

- Proportionally thinner walls than same diameter artery
  - tunica media less muscle
  - lack external & internal elastic lamina
- Still adaptable to variations in volume & pressure
- Valves are thin folds of tunica interna designed to prevent backflow
- Venous sinus has no muscle at all
  - coronary sinus or dural venous sinuses



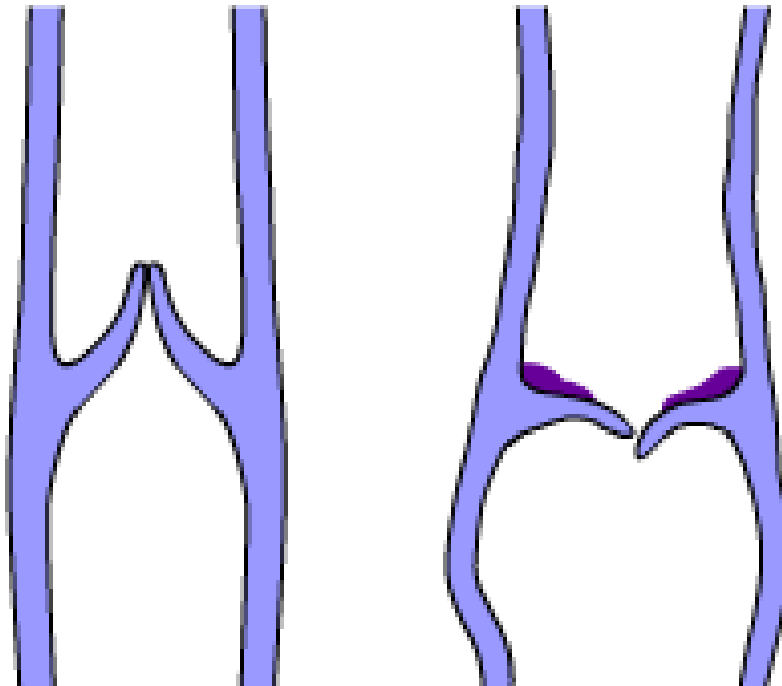
# Venous valves



# Varicose Veins

- Twisted, dilated superficial veins
  - caused by leaky venous valves
    - congenital or mechanically stressed from prolonged standing or pregnancy
  - allow backflow and pooling of blood
    - extra pressure forces fluids into surrounding tissues
    - nearby tissue is inflamed and tender
- Deeper veins not susceptible because of support of surrounding muscles

# Varicose veins

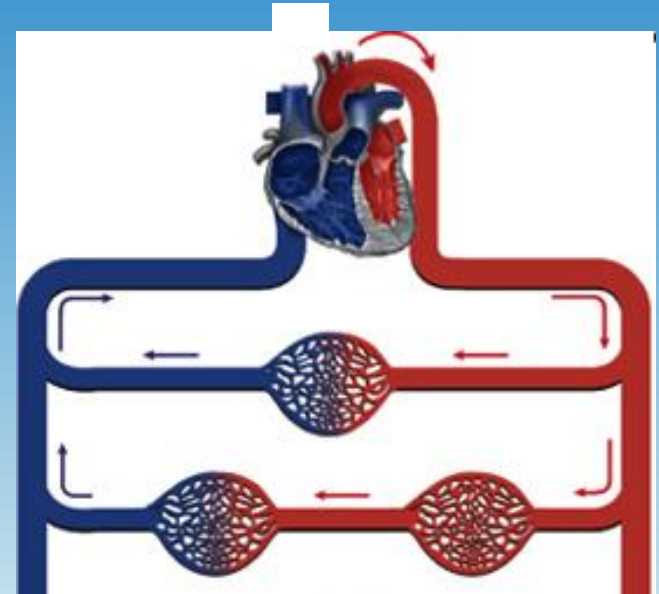


Normal vein  
with healthy valve

Varicose vein  
with leaky valve

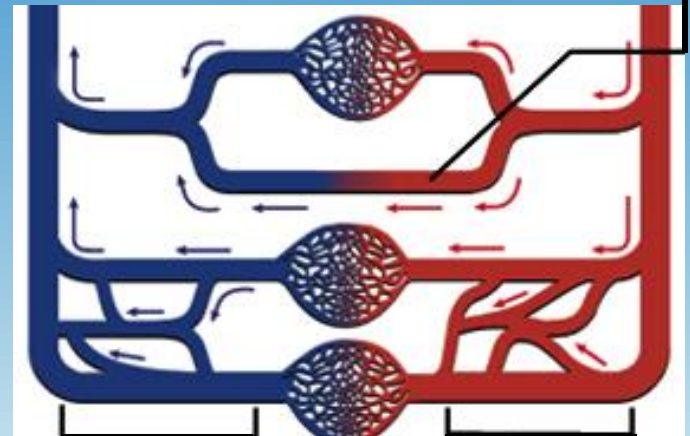
# Circulatory Routes

- Most common route
  - heart → arteries → arterioles → capillaries → venules → veins
- Portal system
  - blood flows through two consecutive capillary networks before returning to heart
    - hypothalamus - anterior pituitary
    - found in kidneys
    - between intestines - liver



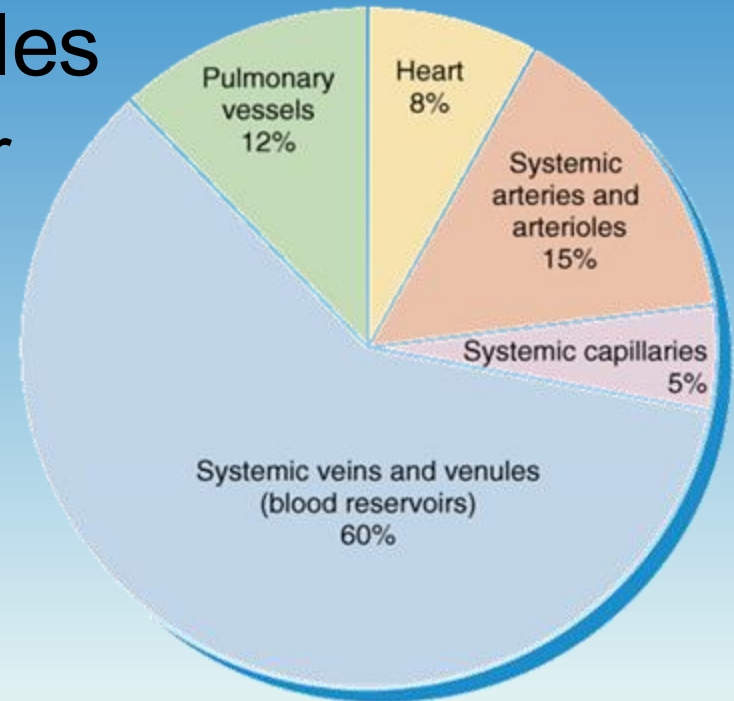
# Circulation Routes: Anastomoses

- Anastomosis = point where 2 blood vessels merge
- Arteriovenous shunt
  - artery flows directly into vein
  - fingers, toes, ears; ↓ heat loss, allows blood to bypass exposed areas during cold
- Venous anastomosis
  - most common, blockage less serious
  - alternate drainage of organs
- Arterial anastomosis
  - collateral circulation (coronary)



# Blood Distribution

- 60% of blood volume at rest is in systemic veins and venules
  - function as blood reservoir
    - veins of skin & abdominal organs
  - blood is diverted from it in times of need
    - increased muscular activity produces venoconstriction
    - hemorrhage causes venoconstriction to help maintain blood pressure
- 15% of blood volume in arteries & arterioles



# Capillary Exchange

- Only occurs across capillary walls between blood and surrounding tissues
- 3 routes across endothelial cells
  - intercellular clefts
  - fenestrations
  - through cytoplasm
- Mechanisms involved
  - diffusion, transcytosis, filtration and reabsorption

# Capillary Exchange

- Movement of materials in & out of a capillary
  - diffusion (most important method)
    - substances move down concentration gradient
    - all plasma solutes except large proteins pass freely across
      - through lipid bilayer, fenestrations or intercellular clefts
      - blood brain barrier does not allow diffusion of water-soluble materials (nonfenestrated epithelium with tight junctions)
  - transcytosis
    - passage of material across endothelium in tiny vesicles by endocytosis and exocytosis
      - large, lipid-insoluble molecules such as insulin or maternal antibodies passing through placental circulation to fetus
  - bulk flow see next slide

# Bulk Flow: Filtration & Reabsorption

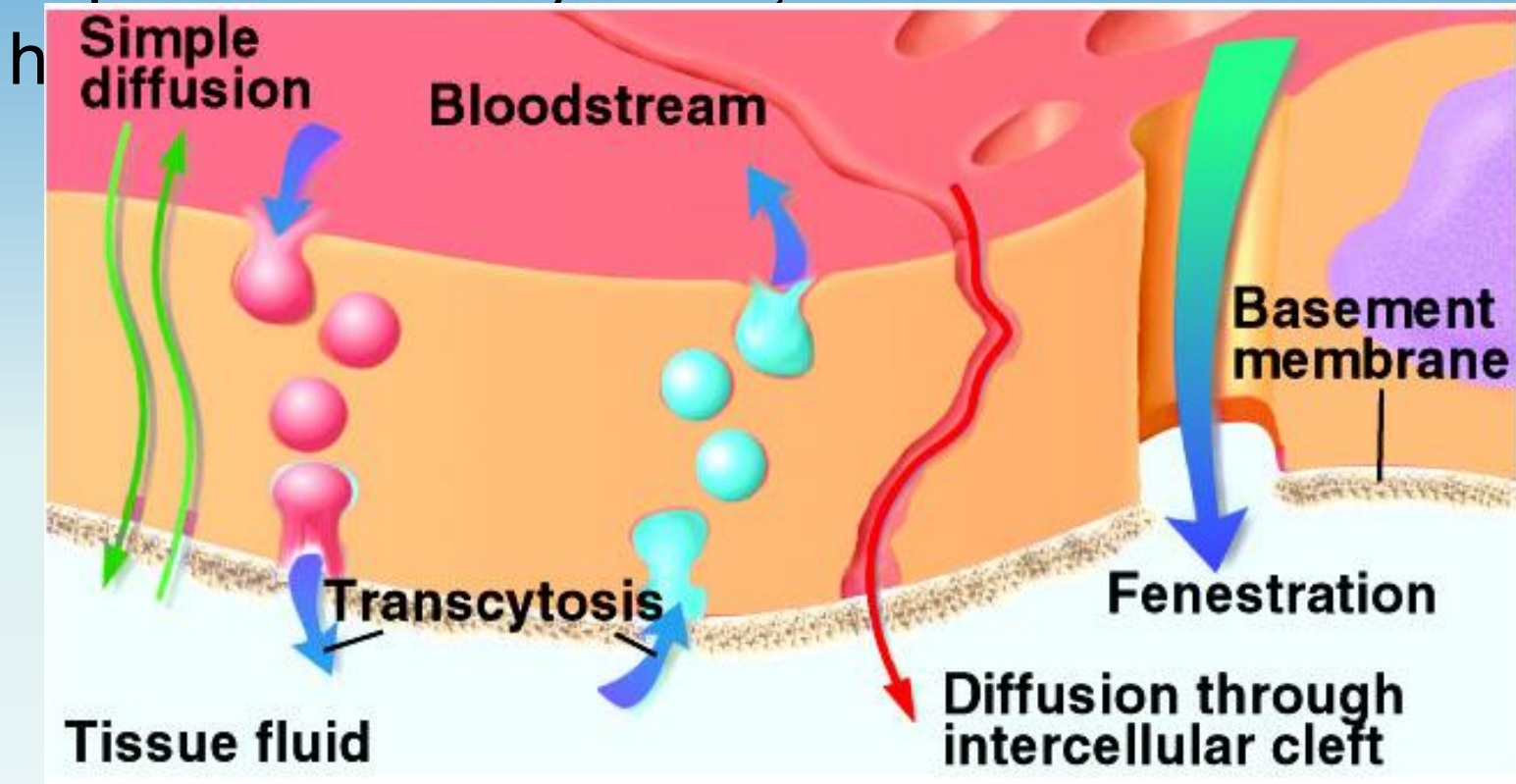
- Movement of large amount of dissolved or suspended material in same direction
  - move in response to pressure
    - from area of high pressure to area of low
  - faster rate of movement than diffusion or osmosis
- Most important for regulation of relative volumes of blood & interstitial fluid
  - filtration is movement of material into interstitial fluid
    - promoted by blood hydrostatic pressure & interstitial fluid osmotic pressure
  - reabsorption is movement from interstitial fluid into capillaries
    - promoted by blood colloid osmotic pressure
  - balance of these pressures is net filtration pressure

# Capillary Exchange - Diffusion

- Most important mechanism
- Lipid soluble substances
  - steroid hormones,  $O_2$  and  $CO_2$  **diffuse easily**
- Insoluble substances
  - glucose and electrolytes must pass through channels, fenestrations or intercellular clefts
- Large particles - proteins, held back

# Capillary Exchange - Transcytosis

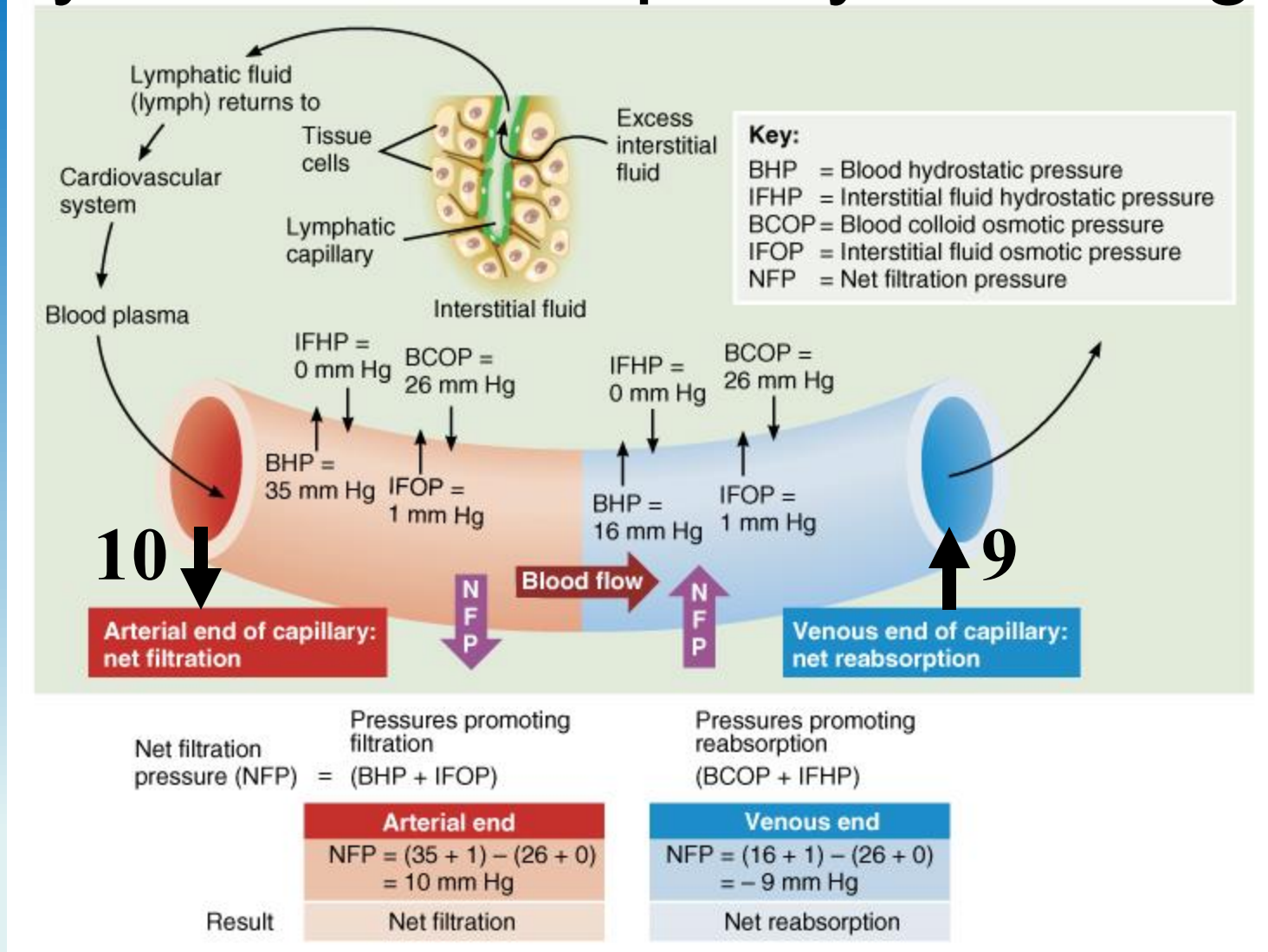
- Pinocytosis, transport vesicles across the cell, exocytosis
- Important for fatty acids, albumin and some



# Capillary Exchange - Filtration and Reabsorption

- Opposing forces
  - blood (hydrostatic) pressure drives fluid out of capillary
    - **high on arterial end** of capillary, **low on venous end**
  - colloid osmotic pressure (COP) draws fluid into capillary (same on both ends)
    - results from plasma proteins (albumin)- more in blood
    - oncotic pressure = net COP (blood COP - tissue COP)

# Dynamics of Capillary Exchange



- Starling's law of the capillaries is that the volume of fluid & solutes reabsorbed is almost as large as the volume filtered

# Net Filtration Pressure

- Whether fluids leave or enter capillaries depends on net balance of pressures
  - net outward pressure of 10 mm Hg at arterial end of a capillary bed
  - net inward pressure of 9 mm Hg at venous end of a capillary bed
- About 85% of the filtered fluid is returned to the capillary
  - escaping fluid and plasma proteins are collected by lymphatic capillaries (3 liters/day)

# Edema

- An abnormal increase in interstitial fluid if filtration exceeds reabsorption
  - result of excess filtration
    - increased blood pressure (hypertension)
    - increased permeability of capillaries allows plasma proteins to escape
  - result of inadequate reabsorption
    - decreased concentration of plasma proteins lowers blood colloid osmotic pressure
      - inadequate synthesis or loss from liver disease, burns, malnutrition or kidney disease
- Not noticeable until 30% above normal

# Edema

- Poor venous return
  - congestive heart failure - pulmonary edema
  - insufficient muscular activity
    - kidney failure (water retention, hypertension)
    - histamine makes capillaries more permeable
- Obstructed lymphatic drainage

# Consequences of Edema

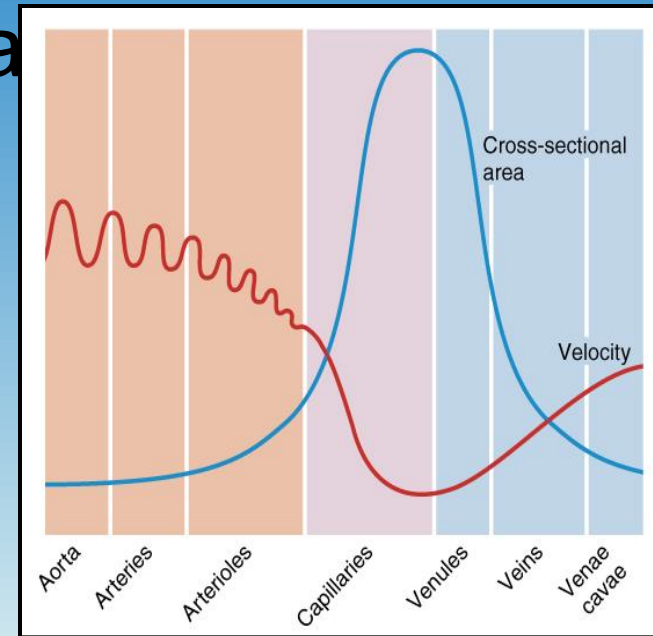
- Circulatory shock
  - excess fluid in tissue spaces causes low blood volume and low BP
- Tissue necrosis
  - oxygen delivery and waste removal impaired
- Pulmonary edema
  - suffocation
- Cerebral edema
  - headaches, nausea, seizures and coma

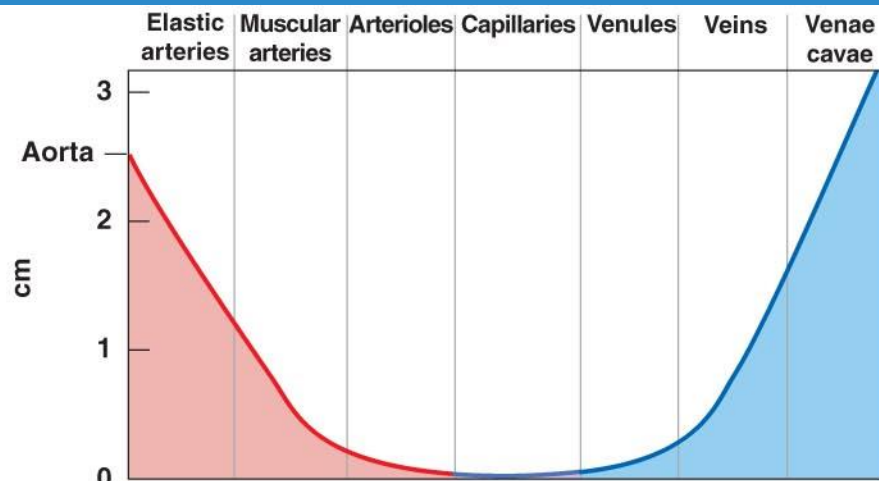
# Hemodynamics

- Factors affecting circulation
  - pressure differences that drive the blood flow
    - velocity of blood flow
    - volume of blood flow
    - blood pressure
  - resistance to flow
  - venous return
- An interplay of forces result in blood flow

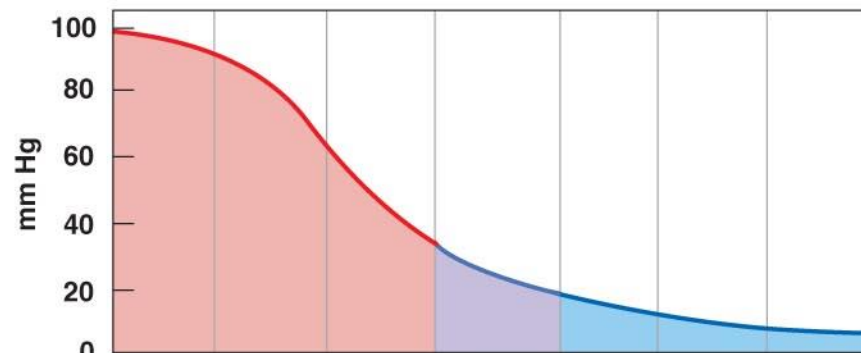
# Velocity of Blood Flow

- Speed of blood flow in cm/sec is inversely related to cross-sectional area
  - blood flow is slower in the arterial branches
    - flow in aorta is 40 cm/sec while flow in capillaries is .1 cm/sec
    - slow rate in capillaries allows for exchange
- Blood flow becomes faster when vessels merge to form veins
- Circulation time is time it takes a drop of blood to travel from right atrium back to right atrium

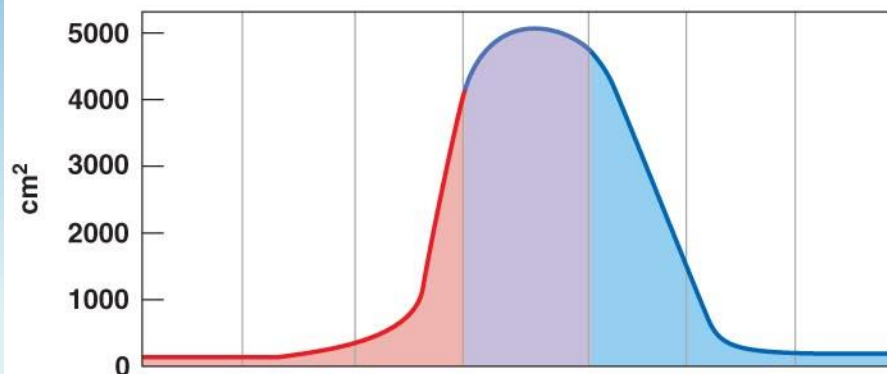




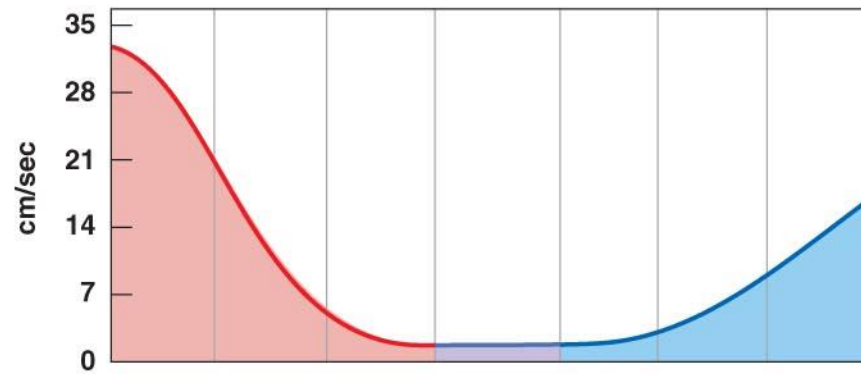
(a) Vessel diameter



(c) Average blood pressure



(b) Total cross-sectional area of vessels



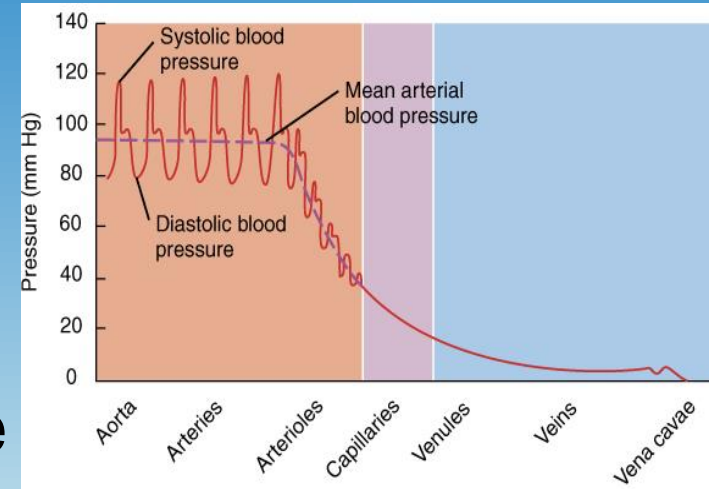
(d) Velocity of blood flow

# Volume of Blood Flow

- Cardiac output = stroke volume x heart rate
- Other factors that influence cardiac output
  - blood pressure
  - resistance due to friction between blood cells and blood vessel walls
    - blood flows from areas of higher pressure to areas of lower pressure

# Blood Pressure

- Pressure exerted by blood on walls of a vessel
  - caused by contraction of the ventricles
  - highest in aorta
    - 120 mm Hg during systole & 80 during diastole
- If heart rate increases cardiac output, BP rises
- Pressure falls steadily in systemic circulation with distance ventricle
  - 35 mm Hg entering the capillaries
  - 0 mm Hg entering the right atrium
- If decrease in blood volume is over 10%, BP drops
- Water retention increases blood pressure



# Abnormalities of Blood Pressure

- Hypertension
  - chronic resting BP > 140/90
  - can weaken small arteries and cause aneurysms

# Stages of Hypertension

- Normal less than 120/80
- Prehypertension 120-129/80-89 either/or
- Hypertension I 130-159/90-99 either/or
- Hypertension II over 160/100 either/or
- Hypertensive Crisis above 180/120 either/or

# Hypotension

- Hypotension
  - chronic low resting BP
  - causes: blood loss, dehydration, anemia
- Orthostatic hypotension

# Mean Arterial pressure

- $MAP = \frac{SBP + 2 (DBP)}{3}$
- Better indicator of perfusion than blood pressure
- Low below 60 = inadequate perfusion
- High above 100 = too much resistance

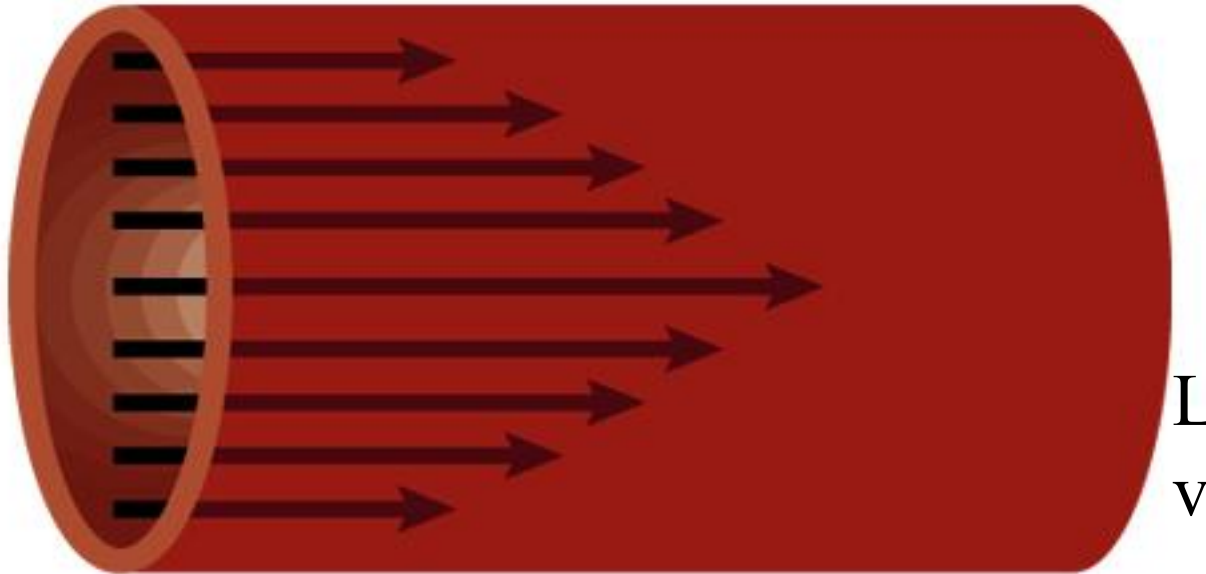
# Blood Pressure 2

- Importance of arterial elasticity
  - expansion and recoil maintains steady flow of blood throughout cardiac cycle, smoothes out pressure fluctuations and ↓ stress on small arteries
- BP rises with age: arteries less distensible
- BP determined by cardiac output, blood volume and peripheral resistance

# Resistance

- Friction between blood and the walls of vessels
  - average blood vessel radius
    - smaller vessels offer more resistance to blood flow
    - cause moment to moment fluctuations in pressure
  - blood viscosity (thickness)
    - ratio of red blood cells to plasma volume
    - increases in viscosity increase resistance
      - dehydration or polycythemia
  - total blood vessel length
    - the longer the vessel, the greater the resistance to flow
    - 200 miles of blood vessels for every pound of fat
      - obesity causes high blood pressure
- Systemic vascular resistance is the total of above
  - arterioles control BP by changing diameter

# Laminar Flow and Vessel Radius



Large radius = average velocity of flow is high

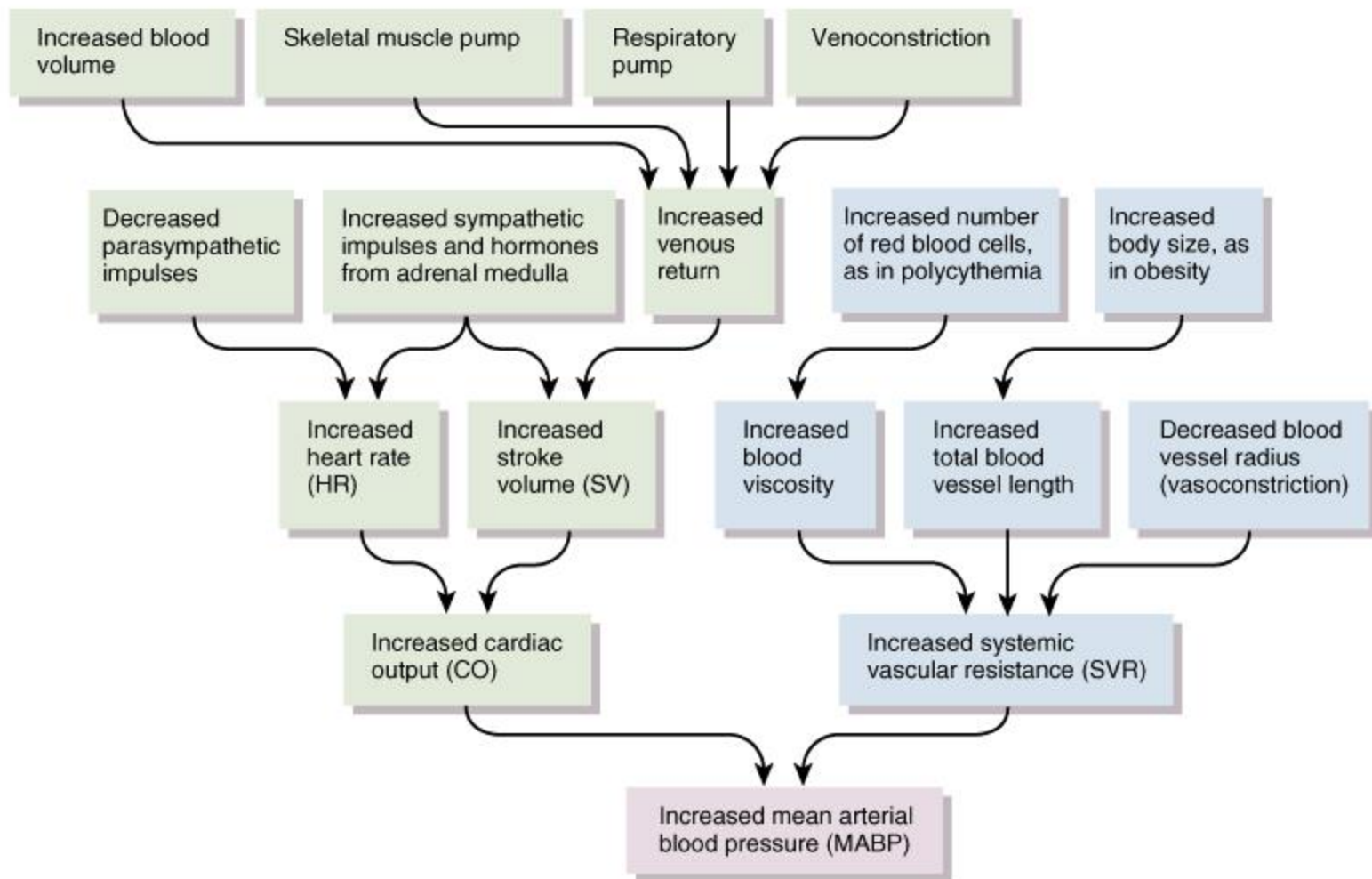


Small radius = average velocity of flow is low

# Resistance

- Vessel radius
  - laminar flow - flows in layers, faster in center
  - blood flow (F) proportional to the fourth power of radius (r),  $F \propto r^4$ 
    - arterioles can constrict to 1/3 of fully relaxed radius
    - if  $r = 3 \text{ mm}$ ,  $F = (3^4) = 81 \text{ mm/sec}$ ; if  $r = 1 \text{ mm}$ ,  $F = 1 \text{ mm/sec}$

# Factors that Increase Blood Pressure



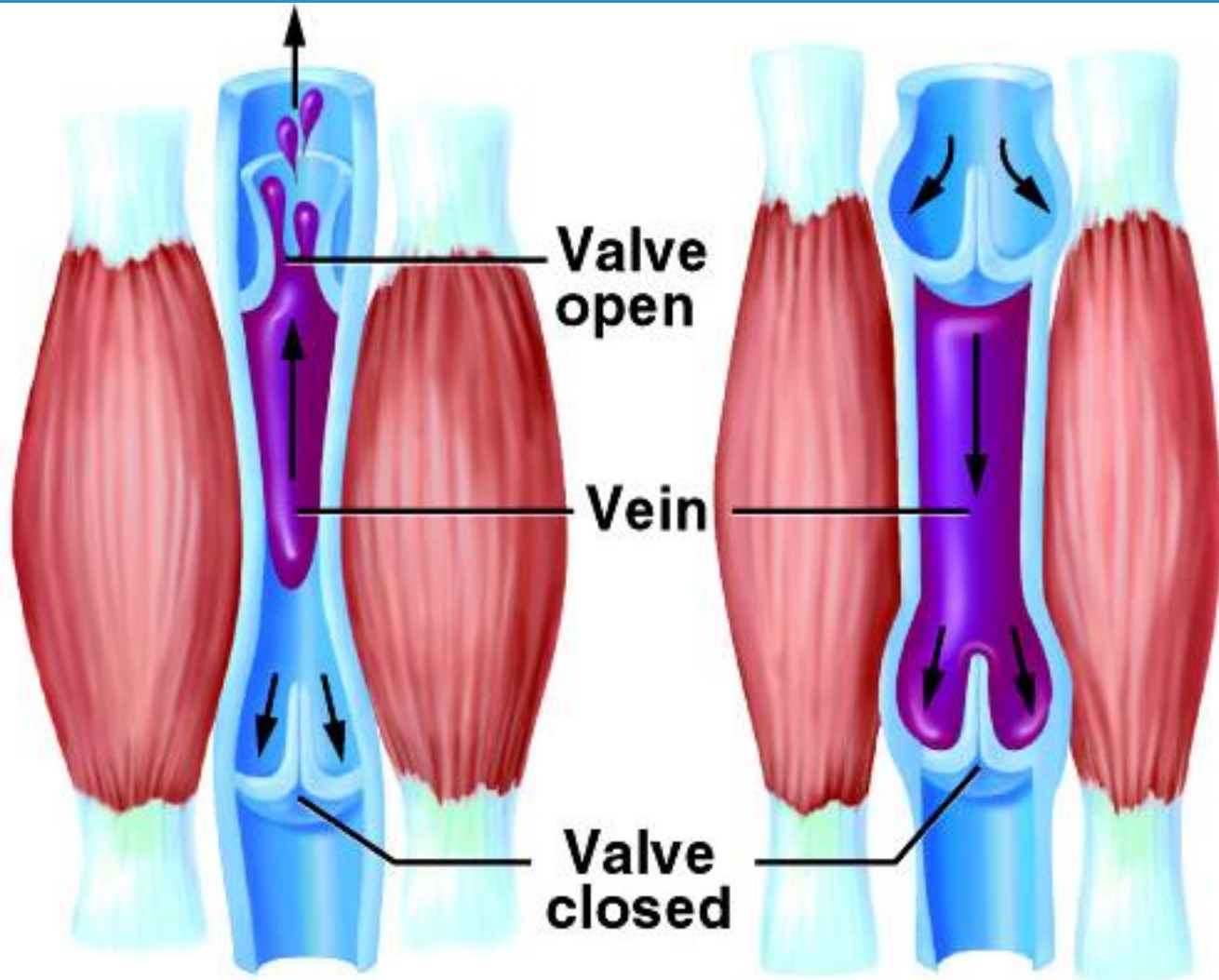
# Flow at Different Points

- From aorta to capillaries, flow ↓ for 3 reasons
  - greater distance traveled, more friction to ↓ flow
  - smaller radii of arterioles and capillaries
  - farther from the heart, greater the total cross sectional area
- From capillaries to vena cava, flow ↑ again
  - large amount of blood forced into smaller channels
  - never regains velocity of large arteries

# Mechanisms of Venous Return

- Pressure gradient
  - 7-13 mm Hg venous pressure towards heart
    - venules (12-18 mm Hg) to central venous pressure (~5 mm Hg)
- Gravity drains blood from head and neck
- Skeletal muscle pump in the limbs
- Thoracic pump
  - inhalation - thoracic cavity expands (pressure ↓) abdominal pressure ↑, forcing blood upward
  - central venous pressure fluctuates
    - 2mmHg- inhalation, 6mmHg-exhalation
    - blood flows faster with inhalation
- Cardiac suction of expanding atrial space

# Skeletal Muscle Pump



# Venous Return and Physical Activity

- Exercise  $\uparrow$  venous return in many ways
  - heart beats faster, harder -  $\uparrow$  CO and BP
  - vessels of skeletal muscles, lungs and heart dilate  $\uparrow$  flow
  - $\uparrow$  respiratory rate  $\uparrow$  action of thoracic pump
  - $\uparrow$  skeletal muscle pump
- Venous pooling occurs with inactivity
  - venous pressure not enough force blood upward
  - with prolonged standing, CO may be low enough to cause dizziness or syncope
    - prevented by tensing leg muscles, activate skeletal m. pump
  - jet pilots wear pressure suits

# Shock and Homeostasis

- Shock is failure of cardiovascular system to deliver enough O<sub>2</sub> and nutrients
  - inadequate perfusion
  - cells forced to switch to anaerobic respiration
  - lactic acid builds up
  - cells and tissues become damaged & die

# Stages of Shock

| Compensated shock                                                                                                                                                                                                                                                                                      | Decompensated shock                                                                                                                                                                                                                                                                | Irreversible shock                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>- Normal blood pressure</li><li>- Tachycardia</li><li>- Redistribution of blood to pivotal organs; brain, heart, adrenal glands</li><li>- Delayed peripheral blood perfusion</li><li>- Decreased urine output</li><li>- May deteriorate if not treated</li></ul> | <ul style="list-style-type: none"><li>- Anaerobic energy production of tissues</li><li>- Worsening metabolic acidosis</li><li>- Renal failure</li><li>- Decreased response to inotropes</li><li>- Hypotension occurs</li><li>- Rapidly deteriorate to irreversible shock</li></ul> | <ul style="list-style-type: none"><li>- Multiorgan dysfunction</li><li>- Poor response to inotropes</li><li>- Intractable metabolic acidosis</li><li>- Death</li></ul> |

# Circulatory Shock

- Any state where cardiac output insufficient to meet metabolic needs
  - cardiogenic shock - inadequate pumping of heart (MI)
  - low venous return (LVR) shock - 3 principle forms
- LVR shock
  - hypovolemic shock - most common
    - loss of blood volume: trauma, bleeding, burns, dehydration
  - obstructed venous return shock - tumor or aneurysm
  - next slide

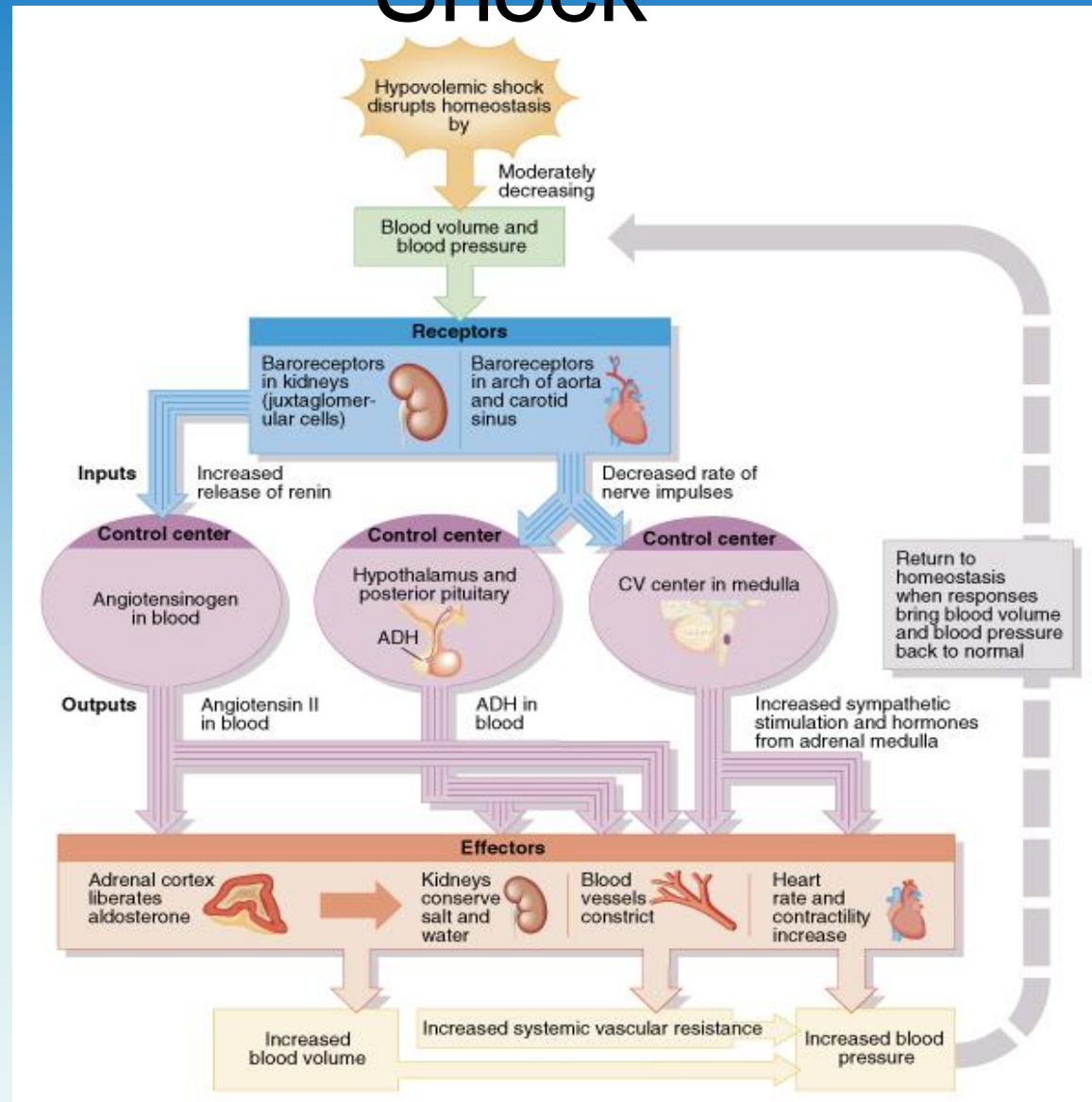
# LVR Shock 2

- Venous pooling (vascular) shock
  - long periods of standing, sitting or widespread vasodilation
  - neurogenic shock - loss of vasomotor tone, vasodilation
    - causes from emotional shock to brainstem injury
- Septic shock
  - bacterial toxins trigger vasodilation and ↑ capillary permeability
- Anaphylactic shock
  - severe immune reaction to antigen, histamine release, generalized vasodilation, ↑ capillary permeability

# Responses to Circulatory Shock

- Compensated shock
  - Homeostatic Mechanisms of compensation in shock attempt to return cardiac output & BP to normal
  - activation of renin-angiotensin-aldosterone
  - secretion of antidiuretic hormone
  - activation of sympathetic nervous system
  - baroreflex
  - release of local vasoconstrictors
  - if person faints and falls to horizontal position, gravity restores blood flow to brain; quicker if feet are raised
- If blood volume drops by 10-20% or if BP does not rise sufficiently, perfusion may be inadequate -- cells start to die

# Restoring BP during Hypovolemic Shock



# Responses to Shock 2

- Decompensated shock  
(life threatening positive feedback loops occur)
  - $\downarrow$  CO  $\rightarrow$  myocardial ischemia and infarction  $\rightarrow$   
 $\downarrow$  CO
  - slow circulation  $\rightarrow$  disseminated intravascular  
coagulation  $\rightarrow$  slow circulation
  - ischemia and acidosis of brainstem  $\rightarrow$   $\downarrow$   
vasomotor tone, vasodilation  $\rightarrow$   $\downarrow$  CO  $\rightarrow$   
ischemia and acidosis of brainstem

# Signs & Symptoms of Shock

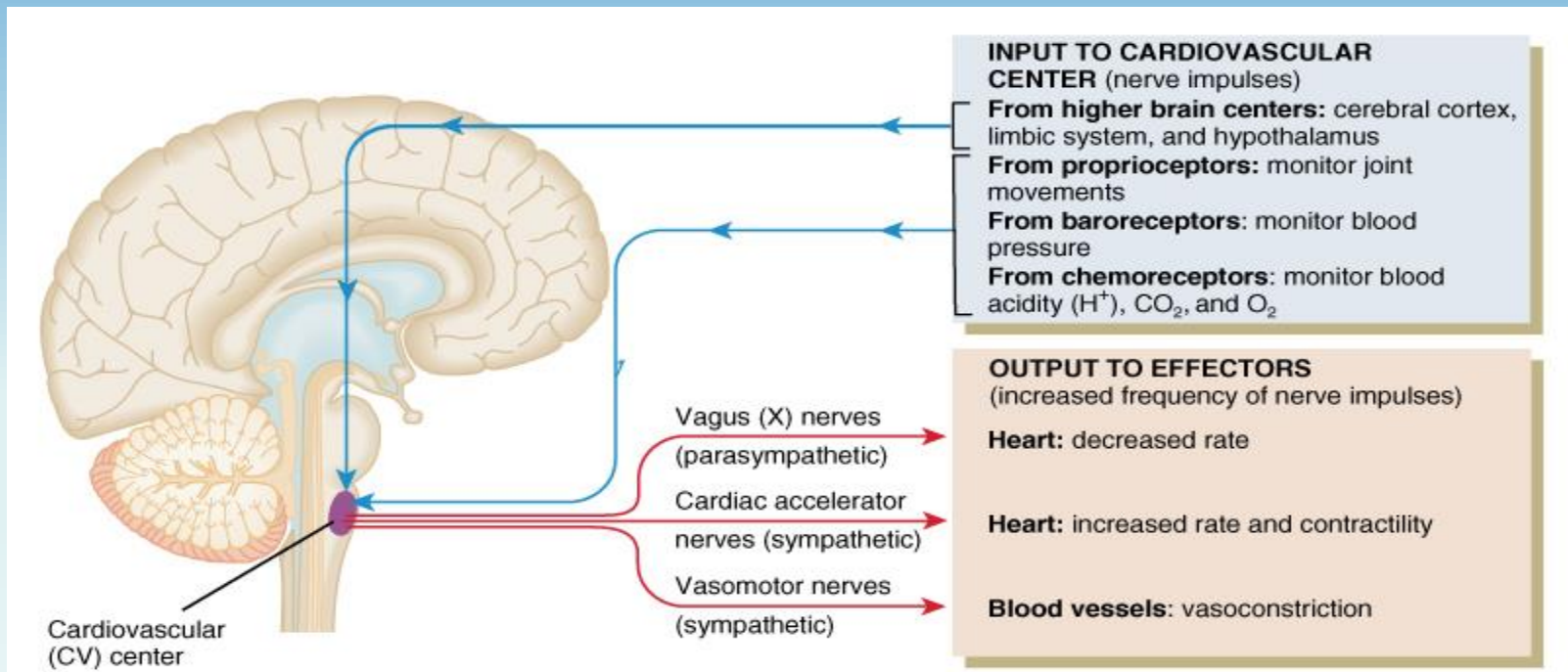
- Rapid resting heart rate (sympathetic stimulation)
- Weak, rapid pulse due to reduced cardiac output & fast heart rate
- Clammy, cool skin due to cutaneous vasoconstriction
- Sweating -- sympathetic stimulation
- Altered mental state due to cerebral ischemia
- Reduced urine formation -- vasoconstriction to kidneys & increased aldosterone & antidiuretic hormone
- Thirst -- loss of extracellular fluid
- Acidosis -- buildup of lactic acid
- Nausea -- impaired circulation to GI tract

# Syncope

- Fainting or a sudden, temporary loss of consciousness not due to trauma
  - due to cerebral ischemia or lack of blood flow to the brain
- Causes
  - vasodepressor syncope = sudden emotional stress
  - situational syncope = pressure stress of coughing, defecation, or urination
  - drug-induced syncope = antihypertensives, diuretics, vasodilators and tranquilizers
  - orthostatic hypotension = decrease in BP upon standing

# Control of Blood Pressure & Flow

- Role of cardiovascular center
  - help regulate heart rate & stroke volume
  - specific neurons regulate blood vessel diameter



# Input to the Cardiovascular Center

- Higher brain centers such as cerebral cortex, limbic system & hypothalamus
  - anticipation of competition
  - increase in body temperature
- Proprioceptors
  - input during physical activity
- Baroreceptors
  - changes in pressure within blood vessels
- Chemoreceptors
  - monitor concentration of chemicals in the blood

# Output from the Cardiovascular Center

- Heart
  - parasympathetic (vagus nerve)
    - decrease heart rate
  - sympathetic (cardiac accelerator nerves)
    - cause increase or decrease in contractility & rate
- Blood vessels
  - sympathetic vasomotor nerves
    - continual stimulation to arterioles in skin & abdominal viscera producing vasoconstriction (vasomotor tone)
    - increased stimulation produces constriction & increased BP

# Neural Control of BP and Flow

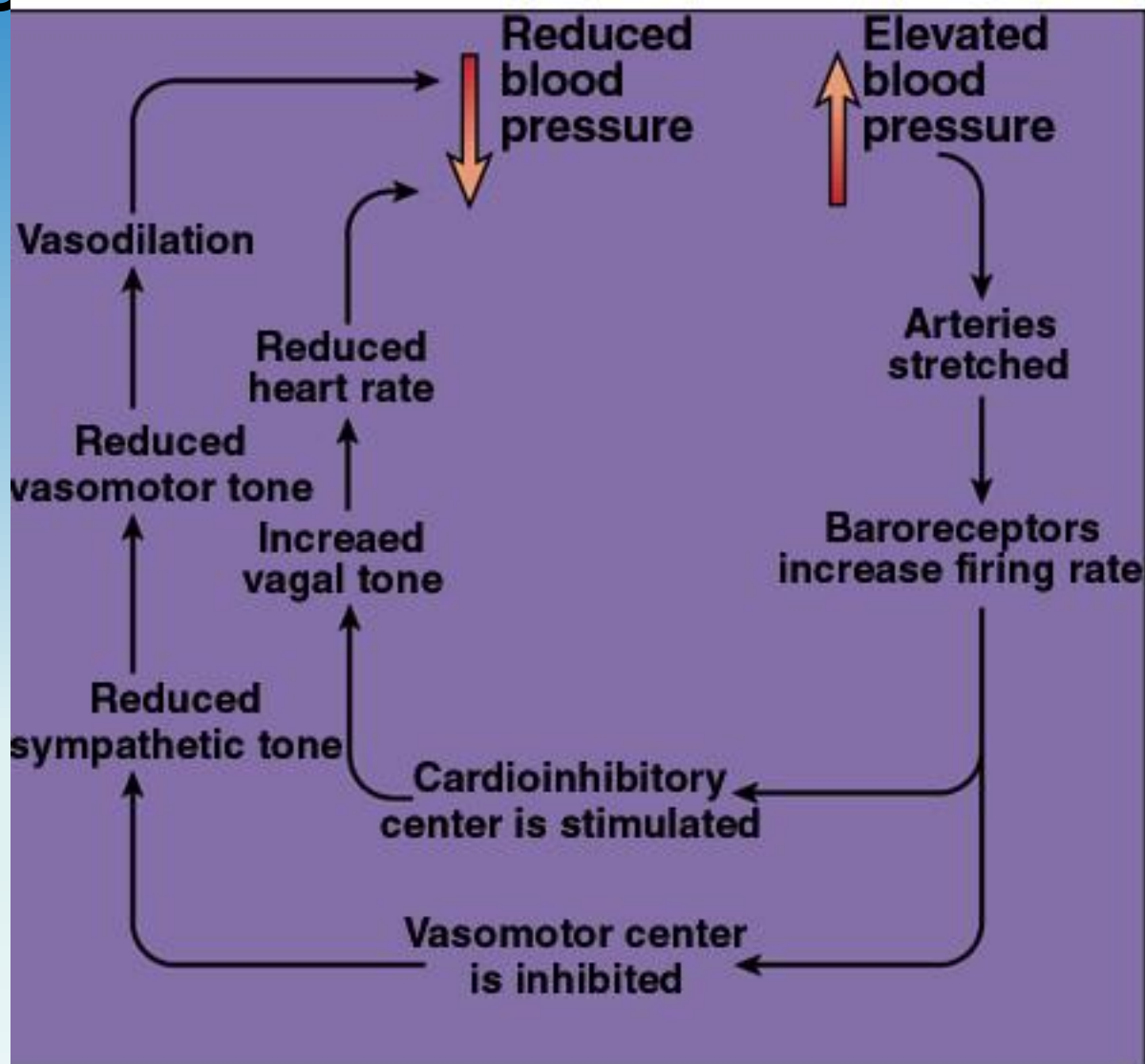
- Vasomotor center of medulla oblongata:
  - sympathetic control stimulates most vessels to constrict, but dilates vessels in skeletal and cardiac muscle
  - integrates three autonomic reflexes
    - baroreflexes
    - chemoreflexes
    - medullary ischemic reflex

# Neural Control: Baroreflex

- Changes in BP detected by stretch receptors (baroreceptors), in large arteries above heart
  - aortic arch
  - aortic sinuses (behind aortic valve cusps)
  - carotid sinus (base of each internal carotid artery)
- Autonomic **negative** feedback response
  - baroreceptors send constant signals to brainstem
  - $\uparrow$  **BP** causes rate of signals to rise, **inhibits** vasomotor center,  $\downarrow$  sympathetic tone, vasodilation causes BP  $\downarrow$
  - $\downarrow$  **BP** causes rate of signals to drop, **excites** vasomotor center,  $\uparrow$  sympathetic tone, vasoconstriction and BP  $\uparrow$

# Baroreflex

## Negative Feedback Response



# Carotid Sinus Massage & Syncope

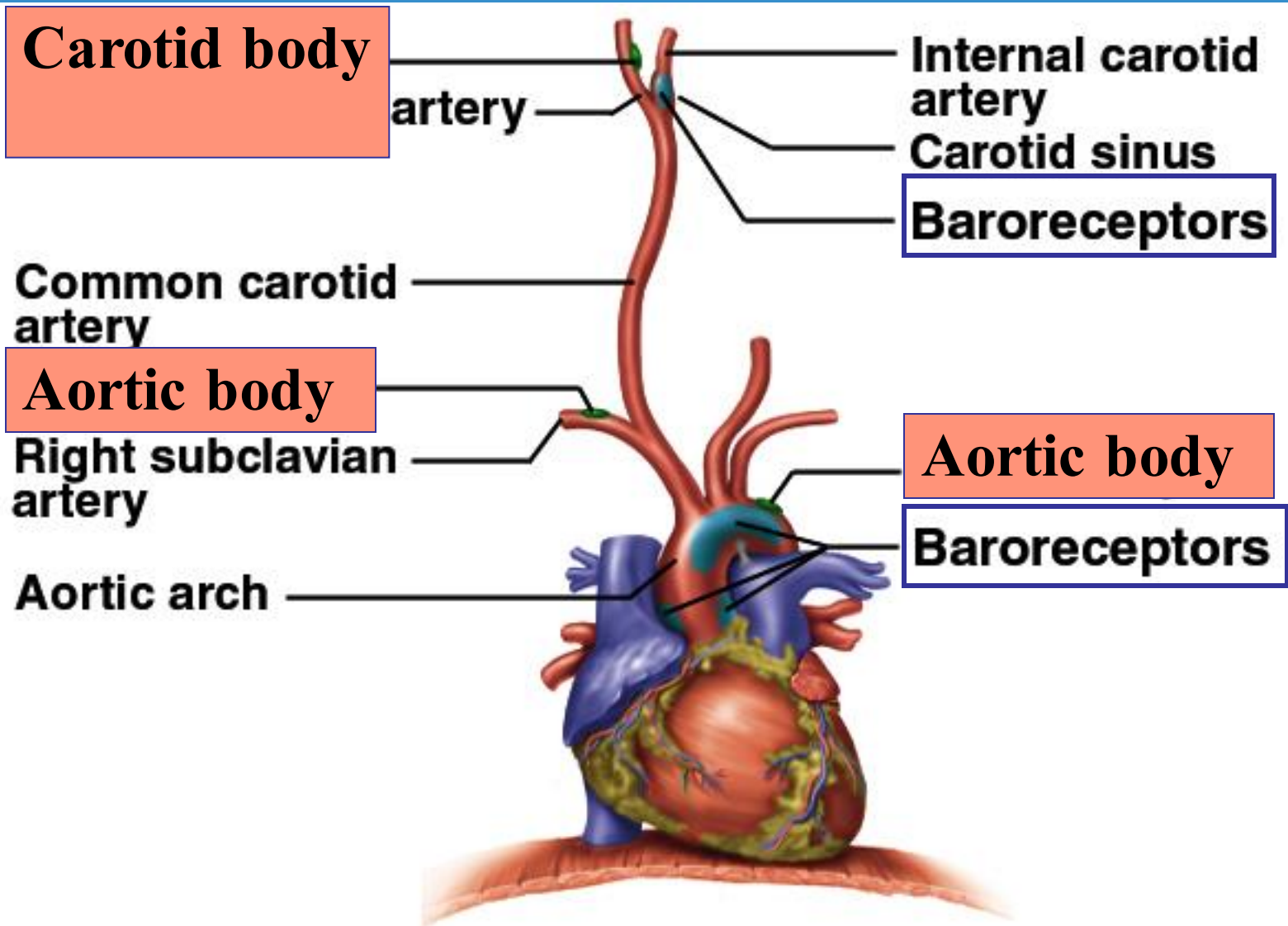
- Stimulation (careful neck massage) over the carotid sinus to lower heart rate
  - paroxysmal supraventricular tachycardia
    - tachycardia originating from the atria
- Anything that puts pressure on carotid sinus
  - tight collar or hyperextension of the neck
  - may slow heart rate & cause carotid sinus syncope or fainting

# Neural Control: Chemoreflex

- Chemoreceptors in aortic bodies and carotid bodies
  - located in aortic arch, subclavian arteries, external carotid arteries
- Autonomic response to changes in blood chemistry
  - pH, O<sub>2</sub>, CO<sub>2</sub>
  - primary role: adjust respiration
  - secondary role: vasomotion
    - hypoxemia, hypercapnia and acidosis stimulate chemoreceptors, instruct vasomotor center to cause vasoconstriction, ↑ BP, ↑ lung perfusion and gas exchange

# Chemoreceptors &

# Baroreceptors



# Other Inputs to Vasomotor Center

- Medullary ischemic reflex
  - inadequate perfusion of brainstem
    - cardiac and vasomotor centers send sympathetic signals to heart and blood vessels
    - ↑ cardiac output and causes widespread vasoconstriction
    - ↑ BP
- Other brain centers
  - stress, anger, arousal can also ↑ BP

# Hormonal Control of BP and Flow

## Angiotensin II

- **Angiotensinogen** (prohormone produced by liver)



Renin (kidney enzyme - low BP)

- **Angiotensin I**



ACE (angiotensin-converting enzyme in lungs)

ACE inhibitors block this enzyme  
lowering BP

- **Angiotensin II**
  - very potent vasoconstrictor

# Hormonal Control of BP and Flow 2

- Aldosterone
  - promotes  $\text{Na}^+$  and water retention by the kidneys
  - increases blood volume and pressure
- Atrial natriuretic factor ( $\uparrow$  urinary sodium excretion)
  - generalized vasodilation
- ADH (water retention)
  - pathologically high concentrations, vasoconstriction
- Epinephrine and norepinephrine effects
  - most blood vessels
    - binds to  $\alpha$ -adrenergic receptors, **vasoconstriction**
  - skeletal and cardiac muscle blood vessels
    - binds to  $\beta$ -adrenergic receptors, **vasodilation**

# Local Regulation of Blood Pressure

- Local factors cause changes in each capillary bed
  - autoregulation is ability to make these changes as needed by demand for O<sub>2</sub> & waste removal
  - important for tissues that have major increases in activity (brain, cardiac & skeletal muscle)
- Local changes in response to physical changes
  - warming & decrease in vascular stretching promotes vasodilation
- Vasoactive substances released from cells alter vessel diameter (K<sup>+</sup>, H<sup>+</sup>, lactic acid, nitric oxide)
  - systemic vessels dilate in response to low levels of O<sub>2</sub>
  - pulmonary vessels constrict in response to low levels of O<sub>2</sub>

# Special Circulatory Routes - Brain

- Total perfusion kept constant
  - few seconds of deprivation causes loss of consciousness
  - 4-5 minutes causes irreversible brain damage
  - flow can be shifted from one active region to another
- Responds to changes in BP and chemistry
  - cerebral arteries: dilate as BP ↓, constrict as BP rises
  - main chemical stimulus: pH
    - $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{H}^+ + (\text{HCO}_3)^-$
    - if  $\text{CO}_2 \uparrow$  (hypercapnia) in brain, pH ↓, triggers vasodilation
    - hypocapnia ↑ pH, vasoconstriction, occurs with hyperventilation, may lead to ischemia, dizziness and sometimes syncope

# Special Circulatory Routes - Skeletal Muscle

- Highly variable flow
- At rest
  - arterioles constrict, total flow about 1L/min
- During exercise
  - arterioles dilate in response to epinephrine and sympathetic nerves
  - precapillary sphincters dilate due to lactic acid, CO<sub>2</sub>
  - blood flow can increase 20 fold
- Muscular contraction impedes flow
  - isometric contraction causes fatigue faster than isotonic

# Special Circulatory Routes - Lungs

- Low pulmonary blood pressure
  - flow slower, more time for gas exchange
  - capillary fluid absorption
    - oncotic pressure overrides hydrostatic pressure
- Unique response to hypoxia
  - pulmonary arteries constrict, redirects flow to better ventilated region

# Aging and the Cardiovascular System

- General changes associated with aging
  - decreased compliance of aorta
  - reduction in cardiac muscle fiber size
  - reduced cardiac output & maximum heart rate
  - increase in systolic pressure
- Total cholesterol & LDL increases, HDL decreases
- Congestive heart failure, coronary artery disease and atherosclerosis more likely